PREDICTING RECIDIVISM THROUGH MACHINE LEARNING

by

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To my family.

PREDICTING RECIDIVISM THROUGH MACHINE LEARNING

by

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The current study attempts to explore more precise solutions for the prediction problem in the U.S. criminal justice system. Using data obtained from the Bureau of Justice Statistics (BJS), *Recidivism of Prisoners Released in 1994*, this project investigates a number of statistical models in terms of their classification performances. Specifically, after a careful feature selection process, the conventional logistic regression model is compared to several machine learning models, including random forests, support vector machines, XGBoost, neural networks, and Search algorithm. According to the results, XGBoost and neural networks outperformed all other models in the comparison. Implications for the U.S. criminal justice system are discussed.

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CHAPTER 1

INTRODUCTION

"Our criminal justice system isn't as smart as it should be. It's not keeping us as safe as it should be. It is not as fair as it should be. Mass incarceration makes our country worse off, and we need to do something about it." – President Barack Obama, July 14, 2015.

Statement of the Problem

According to the most recent Bureau of Justice Statistics (BJS) report, 67.8 percent of the state prisoners released in 2005 in 30 states were arrested within three years of release, while 76.6 percent of these released prisoners were arrested within 5 years of release. Overall, 16.1 percent of released prisoners were responsible for almost half of about 1.2 million arrests within a five-year time window (Durose, Cooper, & Snyder, 2014).

While overall recidivism rates hover around 70 percent, these rates can change depending on the offense type. Within 5 years of release, 82.1 percent of property offenders, 76.9 percent of drug offenders, 73.6 percent of public order offenders, and 71.3 percent of violent offenders were arrested for a new crime. The situation seems a little better for federal offenders. About one-half of federal offenders released in 2005 were rearrested for a new crime or rearrested for a violation of supervision conditions (U.S. Sentencing Commission, 2016).

Recidivism rates inform decision-makers especially in terms of deterrence, reentry malfunctioning, and other social problems. They can also be used as a metric for the performance of correctional facilities (Urahn, 2011). In general, the aforementioned statistics warn about three main problems in the criminal justice system. First, the system fails to deter

individuals who have already been introduced to the criminal justice system. A more effective criminal justice system needs to be designed for the sake of general public safety.

Second, the criminal justice system seems to fail in effectively isolating supervised offenders (probation or parole) from criminal lifestyles. In other words, the recidivism problem might stem from a failure in removing the link ex-prisoners have with their previous criminogenic environment that initiated their criminal careers in the first place. This highlights a reentry problem, the mechanism of which should be designed to focus on effectively supervising ex-offenders in communities they are released into. However, scarcity of resources dictates a more efficient allocation of these resources with respect to which offenders should receive more supervision. Defined in this way, the issue becomes a prediction problem in community supervision. Unfortunately, this mechanism is not functioning as desired since so many ex-offenders return to prison.

Third, there is a small group of offenders released from prison who are responsible for a large volume of crimes. These career criminals seem to have continued their criminal lifestyle from where they left off before their incarceration. Additionally, offenders incarcerated for a violent offense were arrested for another violent crime (33.1%) within five years after their release (Durose et al., 2014). The system may have prevented those crimes if high-risk individuals were identified in both prison and community supervision. Defined in this way, this issue also becomes a prediction problem. The current predictive capabilities (whether clinical or actuarial) are falling short of correctly classifying high-risk offenders.

All of these issues are prediction issues, and indeed they had been long voiced by Tonry some decades ago (1987, p.388). When an offender is introduced to the criminal justice system,

he or she experiences a chain of events, all of which requires some level of decision-making. These decision points include receiving bail, sentencing, security ratings, prison placement, the level of treatment, release decisions (parole or probation), and level of supervision (Kroner & Mills, 2001). The quality of decisions at each of these critical stages will manifest themselves in recidivism rates.

If the accuracy of predictions can be significantly improved, stakeholders may be able to target resources on dangerous offenders, to provide greater leniency to non-dangerous offenders, to reduce incarceration, and thereby to achieve greater crime control at less financial costs (Tonry, 1987). With more accurate predictive tools, the criminal justice system would be better at correctly classifying high-risk offenders and adjusting their stay in prison accordingly. This may prevent future crimes of these offenders, which otherwise may have occurred if they had not been incapacitated. For this reason, identifying these high-risk offenders at stages such as parole decisions and treatment allocation can reduce recidivism rates. In the case of the latter, better prediction improves rehabilitation through reaching higher accuracy in determining the target, the dose, and the duration of the trio of risks-needs-responsivity (Andrews, Bonta, & Hoge, 1990; Berk, 2012).

Similarly, an improved classification can also allow low-risk, non-violent offenders to be released into the community, thereby saving public resources, stabilizing lives of individuals and families, preserving community dynamics. The cost of community supervision is lower than incarceration (Petersillia, 1997), and considering today's high imprisonment rates, community supervision seems to be heavily resorted by criminal justice agencies in the near future. Moreover, a more accurate prediction can be of help by allowing probation officers to adjust

their time and energy on the high-risk offender group (within released population) informed by actuarial analyses. For example, Barnes and Hyatt (2012) note that correctional agencies were able to handle a 28% increase in their overall caseload with a staff that was 15% smaller than it was before the introduction of forecasting tools in Philadelphia.

For these reasons, designing tools that can assess future recidivism risk more accurately than what is already being used is of utmost importance in crime control and public security. The need for better, more accurate predictive models is real and, it is wiser to seek for superior prediction for the criminal justice system (Bushway, 2013). Concerned with these issues, the current study attempts to seek more precise solutions for the prediction problem in the U.S. criminal justice system. Using data obtained from the Bureau of Justice Statistics (BJS), *Recidivism of Prisoners Released in 1994*, this study explores potential improvement in predictive accuracy regarding recidivism. To do so, a number of statistical models were investigated in terms of their performance. Specifically, the conventional logistic regression model is compared to several machine learning models, including random forests, support vector machines, XGBoost, neural networks, and Search algorithm. While some models used are well-known to recidivism researchers, others are more novel tools for recidivism research literature within the field of criminology.

In the following sections, the literature behind recidivism and statistical solutions are discussed in Chapter 2. Descriptive information for the predictors, the rival models to compare, and the overall analytic procedure are presented in Chapter 3. The findings are presented in Chapter 4, and the implications of this research are discussed in Chapter 5.

CHAPTER 2

LITERATURE REVIEW

"*Theories come and go, but fundamental data always remain the same.*"

— Mary Leakey

Recidivism

Recidivism can be defined as the act of re-engaging in criminal offending despite having been punished (Urahn, 2011). There is a long line of research behind recidivism. A number of factors that are found to be associated with recidivism have been documented in various domains, such as prior criminality, demographic factors, employment, education and antisocial tendencies (Blumstein, Cohen, Roth, & Visher, 1986; Gendreau, Little, & Goggin, 1996; Gottfredson & Jarjoura, 1996; Greenwood & Abrahamse, 1982; Pratt & Cullen, 2000; U.S. Sentencing Commission, 2004, 2016). Jurisdictional policies also bear on the amount of recidivism. For example, states with prisons that house more low-risk offenders are less likely to experience recidivism than states with prisons that house more high-risk offenders (Urahn, 2011).

Recidivism has been documented based on offense types as well. Robbery and firearms offenders have overall higher recidivism rates compared to fraud, larceny, or drug trafficking offenders (U.S. Sentencing Commission, 2004). In a study examining 336 homicide offenders, Roberts, Zgoba, and Shahidullah (2007) found that none of these offenders committed another murder, but the highest recidivism for new violent or drug crimes occurred in the felony homicide group (i.e., homicide during the commission of a felony), followed by the altercation precipitated homicide offenders (27%). The domestic violence and accident-related homicide typologies had lower recidivism rates. On the other hand, prisoners sentenced for property

crimes are the most likely to be arrested. Property crimes are followed by drug offenses, public order offenses, and violent offenses (Durose et al., 2014). Sex offenders are less likely to recidivate, and there is some variance in the rearrest rates of different types of sex offenses (Sample & Bray, 2003, 2006).

The following sections describe the effects of basic demographic, criminal history, and some other individual factors as well as a promising intervention method, risk-need-responsivity (RNR) that incorporates those factors.

1. Demographic Factors. Prior research has consistently found race to be a correlate of recidivism. African Americans are more likely than Whites to be rearrested, reconvicted, returned to prison with a new prison sentence, and returned to prison with or without a new prison sentence (Langan & Levin, 2002). Jung, Spjeldnes, and Yamatani (2010) reported that African American males recidivated at a higher rate, and in a shorter time in comparison to White males. In the most recent Bureau of Justice Statistics' report, African Americans had the highest recidivism rate, followed by Hispanics and non-Hispanic Whites within a five-year period (Durose et al., 2014).

However**,** including race as a predictor of future recidivism in the actuarial or clinical analysis is controversial. This issue has ethical and practical dimensions. In terms of ethical concerns, one may argue that race should not be a factor for assessing future dangerousness. Regarding the issue, Berk (2012, p.8) underscored the fact that by excluding race and gender one may be sparing some African-American men substantial time in prison, but at the cost of the deaths of other young African-American men.

In terms of practical issues, the inclusion of race can increase forecasting accuracy. Taxman et al. (2013, p.80) argued that "demographic-neutral" models (that exclude age, gender, ethnic, or racial factors) miss important information in planning an effective strategy against recidivism. Empirical research supports the idea that even after controlling for criminal justice history, demographics of the offender population have an impact on recidivism (Taxman et al., 2013).

On the other hand, Berk (2012) also argued that whatever applies to race can also apply to gender and even to age. Removing these demographic factors will result in substantial loss in performance. Indeed, exclusion of race may not completely remove racial bias in prediction. For example, other predictors commonly used in risk assessments such as zip codes can serve as a proxy for race, especially in racially-segregated areas (Berk, 2009). Similarly, removing race also disallows one to distinguish the independent effect of a given predictor that is correlated with race (Gottfredson & Jarjoura, 1996).

As for gender, males consistently recidivate at higher rates than females (Durose et al., 2014; Langan & Levin, 2002). Among drug offenders, Stahler et al. (2013) found that males were more than twice as likely to be reincarcerated as females. For violent offenders, males are considerably more likely to reoffend violent crimes (Piquero, Jennings, Diamond, & Reingle, 2015).

Age is a predictor for both general recidivism and offense-specific recidivism (Stahler et al., 2013). It is also a powerful contributor to the forecasting performance for violent offenses (Berk & Bleich, 2013). The likelihood of recidivism declines as the prisoner gets older (Langan & Levin, 2002). Durose et al. (2014) reported that 75.9% of inmates who were age 24 or younger

at the time of their release had been arrested for a new offense within three years. The number was 69.7% for those ages 25 to 39 and 60.3% of those ages 40 or older inmates. For federal offenders, released prior to age 21 had the highest rearrest rate, 67.6 percent, while offenders over sixty years old at the time of release had a recidivism rate of 16.0 percent (U.S. Sentencing Commission, 2016). In general, younger males are known to be at risk for violent recidivism (Piquero et al., 2015).

2. Criminal History. The most important factor that is related to recidivism is one's prior criminal record (Blumstein, Cohen, & Farrington, 1988; Piquero, Farrington, & Blumstein, 2003). This is consistently found in many research papers or government reports (Blumstein et al., 1988; Greenwood & Abrahamse, 1982). In the BJS report released in 2002, about half of offenders with three priors were known to recidivate in comparison to about 80 percent of offenders with more than 10 priors in three years (Langan & Levin, 2002). This finding did not change much in the report released in 2014, which indicated that 26.4 percent of inmates with four or fewer prior arrests were rearrested, while 56.1 percent of inmates with 10 or more priors were rearrested in a year after their release (Durose et al., 2014). Following the Sentencing Reform Act of 1984, the U.S. Sentencing Commission heavily relied upon one's past criminal record to assess one's future risk. Criminal history is not just the number of prior arrests one may have. The components of prior criminality, such as offending frequency, seriousness, and recency can also have individual effects on recidivism (Hoffman, 1983).

3. Other individual-level factors. Changes in antisocial attitudes, associates, or personality are associated with a change in recidivism (Serin, Lloyd, Helmus, Derkzen, & Luong, 2013). DeJong (1997) found that individuals with fewer social bonds were less likely to

be deterred by incarceration. Offenders with personality disorders were found to be more likely to be reconvicted (Yang, Liu, & Coid, 2010). In a meta-analytic review, Gendreau, Little, and Goggin (1996) reported the strongest predictors of adult recidivism as criminogenic needs, criminal history/history of antisocial behavior, social achievement, age, gender, race, and family factors. They documented that less robust predictors are intellectual functioning, personal distress factors, and socioeconomic status.

In a study examining 208 sexual offense recidivists and 201 non-recidivists, Hanson and Harris (2000) documented that recidivists were generally considered to have poor social supports, attitudes tolerant of sexual assault, antisocial lifestyles, poor self-management strategies, and difficulties cooperating with supervision. Also, the recidivists showed increased anger and subjective distress just before reoffending. Similarly, Stalans, Yarnold, Seng, Olson, and Repp (2004) reported that the strongest predictor of violent recidivism was whether the offender had generalized aggression. Finally, one of the consistent correlates of crime and delinquency, peer delinquency (Matsueda, 1989; Warr & Stafford, 1991) can also influence recidivism (Benda, 2003; Warr, 1998).

Prior research has consistently found that substance abuse is more prevalent in offender populations than in the general population (Lurigio et al., 2003; Taxman, Perdoni, & Harrison, 2007). Taxman et al. (2013) underscored the importance of distinguishing between lifetime use (ever used), regular use (abuse), and daily life use (dependence) for recidivism. Substance use could increase the likelihood of recidivism among violent offenders (Stahler et al., 2013). Similarly, employment, education, marriage, illicit drug use, and sentencing type were all found to be related to recidivism according to reports (U.S. Sentencing Commission, 2004).

There is not much known about the thinking patterns of criminals. The research that does exist suggests that offenders are known to have errors in thinking, most notably, dominance, entitlement, self-justification, displacing blame, optimistic perceptions of realities, and blaming society (Taxman et al., 2013; Yochelson & Samenow, 1976). This issue is related to specific needs of offenders and addressed in the following sections.

4. The risk-need-responsivity model and recidivism. Devised by Andrews and Bonta (2010), the RNR model intends to reduce recidivism through providing a selective focus on inmates. The risk principle matches the level of service to the level of risk (who to treat), while the need principle makes a distinction between criminogenic needs and non-criminogenic needs and focuses on the criminogenic needs to reduce recidivism (what to treat). The responsivity principle focuses on the way a treatment will be delivered (how to treat) (Andrews & Bonta, 2010).

The RNR model for correctional systems emphasizes the importance of classification both in risk level and in treatment need to achieve three pillars of correctional goals: recidivism reduction, least restrictive sanctioning, and cost effectiveness (Taxman et al., 2013). The RNR model matches offenders to appropriate supervision levels and services based on their static risks (with no or little change over time) and dynamic criminogenic needs (they change over time) (Taxman et al., 2013).

Static risk factors include the age of the first arrest, the number of prior arrests, the number of prior convictions, the number of escapes or infractions in prison, the number of probation violations, and the number of incarcerations, while dynamic risk factors indicate factors that can change. Dynamic risk factors (such as employment, peer association or substance

use) are categorized as criminogenic needs when they relate to recidivism (Andrews & Bonta, 2010; Taxman et al., 2013). Some reports indicated that the most predictive items for future offending were static or historical items (Coid et al., 2007). Other studies argued that dynamic predictors are as useful as the static predictors in predicting recidivism (Gendreau et al., 1996).

Prediction

A prediction in its essence is an argument about an uncertainty. However, this argument is informed by experience, rather than a simple guess. In the case of the criminal justice system, stakeholders would want to estimate with great accuracy, for example, who will violently reoffend upon release. The idea that classifying high-risk offenders results in a better allocation of resources and safer communities has attracted much attention in the literature (e.g., Greenwood & Abrahamse, 1982). It is also important to note that there are ethical issues with prediction in the criminal justice system. For example, according to the "just deserts" philosophy, the punishment should fit the crime, and individuals should not pay an additional price for potential future offenses that they have not yet committed (based on the group they are classified into). This is the exact ethical dilemma regarding the prediction of future criminality. As Stevenson, Cruikshank, and Moldoveanu (1998) note, predicting things is important, but also hard in many ways.

That said, classification and prediction problems have always been at the heart of criminology (Gottfredson, 1987). One can trace back prediction concerns to the early twentieth century. Warner (1923) argued that only clinical diagnosis can be used in predicting reoffending, rather than factors such as demographic, family, and employment, while Hart (1923) rejected the idea and argued that there could be factors that can be useful in prediction of reoffending.

Another author from the same era, Burgess (1928) designed a risk assessment tool for the Illinois Parole Board in which he focused mostly on the static risk factors. Starting from these early studies, risk assessment tools continued to evolve following different forms in four generations (Andrews, Bonta, & Wormith, 2006; Taxman et al., 2013).

The first generation relied on professional judgment, and it was also highly vulnerable to subjectivity. The first-generation assessments were mostly unstructured professional judgments of the probability of offending behavior and for this reason, they were highly subjective. The second generation was the stage at which actuarial assessment efforts started to appear. Second-generation assessments were atheoretical and empirically-based risk instruments with a focus on static items (Andrews et al., 2006). Third generation assessment was based on theoretically informed tools that incorporated dynamic risk factors. Third generation tools can be seen as improvements over the previous generations due to its more structured assessment. Finally, the fourth generation has focused on treatment matching/case management. That is, risk and needs are calculated to provide the best service with an emphasis on responsivity of the offender for the given treatment. In this generation, service and supervision were organized from intake through case closure (Andrews et al., 2006).

1. Common Risk Assessment Instruments. Since the 1970s, in an effort to build a fair and predictive decision-making mechanism in sentencing, the U.S. Parole Commission has paid considerable attention to a quantified instrument (i.e., the Salient Factor Score (SFS) that can help to achieve this goal (Hoffman & Beck, 1974; Hoffman, 1983, 1994). In 1973, the U.S. Board of Parole adopted the SFS as its main decision-making instrument, and since then the Commission has been using SFS as a predictive tool. Following the Sentencing Reform Act of

1984, the U.S. Sentencing Commission (USSC) developed an empirical recidivism risk prediction instrument, the Criminal History Category (CHC) (U.S. Sentencing Commission., 2005). Both SFS and CHC use information on the frequency, seriousness, and recency of prior offenses, as well as the offender's status under a criminal justice sentence in their structure.

While both SFS and CFC focused heavily on the number and timing of prior criminal history events, however, the computation strategy of the individual scores, the time periods employed, and the assignment of points differ (U.S. Sentencing Commission, 2005). The CHC and SFS have one key philosophical/conceptual difference. The purpose of the SFS is to evaluate the likelihood of recidivism, while the CHC intends to serve beyond recidivism prediction. The CHC is concerned with just punishment and deterrence, and closely related to these, culpability as giving harsher punishments for offenders with aggravated prior criminal backgrounds (U.S. Sentencing Commission, 2005). This clear warning intends to convey a message to society that habitual criminal behavior will be subject to greater punishment as a primary pillar of a general deterrence strategy. While prior assessments reported that SFS has more predictive power than CHC (Hoffman & Beck, 1997), recent examinations by the Commission have shown that CHC can outperform SFS through certain modifications. The CHC is currently the recidivism prediction tool employed at federal guideline criminal sentencings to measure offender culpability, deter criminal conduct, and protect the public from further crimes of the defendant (U.S. Sentencing Commission, 2005).

Federal judges use two axes of a sentencing table to determine the appropriate sentencing range. The vertical axis of this table has various offense levels designed to quantify the seriousness of the offense, while the horizontal axis has criminal history categories designed to

quantify the extent and recency of past criminal behavior. The intersection of the offense level and the criminal history level identifies the minimum and the maximum number of months for a recommended sentence (U.S. Sentencing Commission, 2004). In practice, the Sentencing Commission utilizes the criminal history measure as a tool to determine offender culpability, to deter criminal conduct (more punishment for habitual offenders), and to protect the public from future crimes of the defendant (U.S. Sentencing Commission, 2004).

Among other risk assessment tools used by scholars and practitioners are the Offender Group Reconviction Scale (Copas & Marshall, 1998; Howard, Francis, Soothill, & Humphreys, 2009), the Statistical Information about Recidivism Scale (Nuffield, 1982, 1989), the Statistical Information on Recidivism - Revised 1 (SIR-R1) Scale (Nafekh & Motiuk, 2002), and the Level of Service Inventory-Revisited (LSI-R) (Andrews & Bonta, 2000). There are numerous others that are used in various jurisdictions. These scales or guidelines share the common goal of providing predictive improvement. However, none of these are instruments mentioned in this section utilize machine learning algorithms nor differential cost perspective (details will be discussed in the following sections).

2. Actuarial Analysis and Criminology. The debate regarding the modus operandi of risk assessment centers mostly on the superiority between clinical and actuarial assessment in the criminal justice decision-making process. Clinical assessment is supported on the grounds that expertise opinion can be included in the decision-making process (Harcourt, 2007). However, personal experience comes with subjectivity, bias and all kinds of human judgmental characteristics which might manifest themselves in race, ethnicity, religion, or gender. Furthermore, actuarial assessments can at least be as good as human assessment and

sometimes better (Grove & Meehl, 1996). In addition, actuarial assessment is fairer and most importantly, replicable by others.

Actuarial decision making has been found to be superior to human judgements in various situations, including graduate school admissions (Dawes, 1979), parole board decision and parole violation criterion (Carroll, Wiener, Coates, Galegher, & Alibrio, 1982; Glaser, 1962), and recidivism in general (Holland, Holt, Levi, & Beckett, 1983). The reason behind this improved decision-making could be that individuals may inappropriately weight specific information erroneously and may tend to be greatly influenced by a certain causal association (Gottfredson & Moriarty, 2006). Even available prediction tools are not perfect, actuarially-supported decision making outperforms the one without (Gottfredson & Gottfredson, 1994). The subjective exercise of discretion at extreme levels can result in arbitrary decisions as well as unwarranted disparity (Hoffman, 1994).

Unfortunately, while being superior to clinical judgment, the actuarial methods that have been used so far in criminology have their own issues. First, until recently, most predictive models were tested using the same data that was used to obtain the coefficients for the model parameters, which results in over predicting the model parameters (Berk, 2012). Instead, a predictive model should be validated using external data that have not been used by any means in producing parameter coefficients (Zeng, Ustun, & Cynthia, 2015). Using a test sample allows an external validation about the hypothesized association that was obtained in the training sample and can be generalized to other samples within the same population (Berk, Sherman, Barnes, Kurtz, & Ahlman, 2009). This procedure increases the external validity of the findings obtained. Unfortunately, the use of a test sample has been largely ignored by prior criminological literature

on prediction (Berk & Bleich, 2013). For this reason, risk assessment tools in use should be re-evaluated if they were not previously validated using external data.

Second, most conclusions from various prediction studies in the literature relied on limited sample sizes, which is not a desirable situation for many machine-learning algorithms (Coid et al., 2007; Kroner & Mills, 2001; Yang et al., 2010).

Third, forecasting models should consider the relative costs of false positives and false negatives (Berk, He, & Sorenson, 2005). It is essential to consider how much costlier it would be to erroneously classify a high-risk serious offender as low-risk offender (false negative) than it is to waste resources on someone by erroneously classifying the case as high risk (false positive) when in fact they may not be high risk at all (Barnes & Hyatt, 2012). This important limitation is discussed further in the next section.

Fourth, criminology as a field is lagging behind other branches of science when it comes to incorporating novel algorithmic decision-making tools. This is partly due to having theoretical concerns and avoiding a somewhat "defamed" concept "data mining" in the social sciences. That is not necessarily the most appropriate stance toward prediction problems as explanation and forecasting are two different dimensions with different goals. In prediction, one is not much concerned with providing a causal pathway between predictors and the outcome; rather one wants to use observed patterns to estimate future outcomes. This very characteristic of prediction links prediction directly to the policy because of its potential to reduce the incarcerated population. Correct classification of the outcome of interest is directly related to safety, costs, and efficiency, which are main concerns in designing public a policy. For this reason, forecasting is indeed a policy issue.

3. Evaluating costs. The essential task of forecasting is to maximize forecasting accuracy (Berk & Bleich, 2014), but this goal should be revised by the relative misclassification cost, as Bradley (1997) argued, minimizing misclassification cost is more important than misclassification rate. There are different costs in the U.S. criminal justice system regarding domains, such as police, pretrial services, courts, transportation, jail or prison, probation or parole. Ideally, a high true positive rate (TPR) and a low false positive rate (FPR) are desirable in prediction (Berk & Bleich, 2014). However, it is important to note that there is a tradeoff between false positives and false negatives. In the case of equal treatment, false negatives and false positives may statistically produce a higher accuracy, but in fact, as in most fields, these costs almost never have the same consequences. For example, incorrectly classifying a case as potential credit card fraud in a bank may result in wasting some resources, such as time and personnel that are allocated to investigate the matter. However, incorrectly classifying a fraudulent activity may cost the bank substantial sums of money.

Berk (2011) convincingly argued that when the costs of forecasting errors are asymmetric, and an asymmetric loss function is properly deployed, it is possible to inform policies with greater precision. Forecasts that are overestimated (erroneously classifying many as recidivist) may lead to a waste of resources (spending extra for those who are not likely to reoffend) and to all other unwanted social consequences (all negative effects of prison on the incapacitated and his/her family, loss of workforce, etc.) for incapacitating a person longer than needed. On the other hand, forecasts that are underestimated (erroneously classifying most as non-recidivists) have serious consequences, such as an increase in crime. The latter situation gets

worse in the case of violent offenses, such as a murder or rape that could have been prevented if the case were correctly classified.

For example, generating more false positives than false negatives indicates that more offenders are placed into the high-risk category than the actual amount of high-risk offenders. This situation will increase high-risk accommodation in prisons and caseload size and officer workload in community supervision (Barnes & Hyatt, 2012). Similarly, In the case of domestic violence, false negatives can be more problematic than were false positives. For example, the Los Angeles Sheriff's Department considered not responding to a call when there actually was a need for police as much more costly than servicing for a call that turned out to be a false alarm (Berk et al., 2005). In these cases, researchers and practitioners can agree on a predetermined cost ratio, such as 5:1, which was the case in Berk et al. (2005) study.

The ratio of false positives to false negatives can be considered as a cost ratio, which can indicate relative costs attached to either condition. For example, a cost ratio of 2.5 may mean that the cost of mistakenly classifying is 2.5 times more for false negatives than false positives. There is no hard and fast rule about this cost ratio, instead, it is mostly at practitioners' discretion and may well vary from jurisdiction to jurisdiction (Barnes & Hyatt, 2012; Berk, 2011). Berk (2011) advises that decision-makers should decide on the relative cost of forecasting error, and if no stakeholder is involved, a reasonable (but differential) cost of forecasting error can be suggested. In prior literature, for example, the 10 to 1 and 5 to 1 cost ratios were implemented in the case of predicting prisoner misconduct. The ratio can be adjusted based on the specific research question and specific needs of the stakeholder (Berk, Kriegler, & Baek, 2006). Also, a predefined ratio in a department may be modified over time according to emerging needs of the policy-makers.

4. Evaluating Predictive Performance. In criminology, false positive and false negative rates, as well as total correct predictions, are commonly used as a comparison metric among rival models (Caulkins, Cohen, Gorr, & Wei, 1996). True positive (TP) can be seen as a hit, true negative (TN) as correct rejection, false positive (FP) as a false alarm, and false negative (FN) as missing a positive case that behaved in line with the outcome of interest. Some studies reported their predictive accuracy as being correct nearly two-thirds (66.1%) of the time (Barnes & Hyatt, 2012).

In general, stakeholders may want to see high accuracy, that is, high true positives and true negatives. But for the aforementioned reasons regarding relative costs, a mere accuracy number may not be that illuminating. Statistically, desirable outcomes may not be a perfect match in policy (Berk, 2012).

A more nuanced measure is the use of receiver operating characteristic (ROC) curve, which is commonly used to evaluate the performance of a classifier, rather than a simple accuracy rate. A ROC curve analyses prediction accuracy using a graph that plots the true positive rates against the false positive rates, and the area under the curve depicts the predictive power of the prediction tool (Bradley, 1997). The diagonal line represents no predictive power or random chance in ROC curves. This visual representation must be able to show how often the model accurately predicts recidivism, in comparison to how often the model predicts recidivism incorrectly (U.S. Sentencing Commission, 2004). The statistic used to measure the predictive power of the model is defined as the Area Under the Curve (AUC), and a greater AUC value

represents a better the predictive power of the model.[1] A growing number of studies in the criminological literature (concerned with a prediction task) report AUC values to show how good a job their predictive model did.

Predictive power, in general, has been best presented by how instruments accurately predict future recidivism. In Schwalbe's (2007) review, the overall AUC was .640, and third-generation instruments were more accurate than second-generation instruments, with an average AUC of .646. The statistical power of prediction reaches a maximum around AUC of 0.75 (Kroner & Mills, 2001; Coid et al., 2007). The U.S. Sentencing Commission (2004) reported AUC scores ranging from 0.6786 to 0.6992 with combined contributions of the variables, such as criminal history, gender, age, race, employment, education, marriage, illicit drug use, and sentencing type in their report. In a later report, the Commission obtained an AUC of 0.7313 using Salient Factor Score, and 0.6992 using CHC (U.S. Sentencing Commission, 2005). However, when the age element and the first offender element were included in CHC, the resulting AUC value was 0.7462.

Unfortunately, not many studies report their AUC values using a test sample or cross-validation procedures. A training sample error is, by definition, expected to be lower than a testing sample error. For this reason, reporting AUC (using external data that validate the findings) is preferred in recent studies. Among the studies using cross-validation technique, McKinlay, James, and Grace (2015) reported an AUC of 0.71. Liu et al. (2011) reported an

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[1] A classifier that performs no better than chance has AUC of 0.5, and anything above 0.5 is an improvement over chance (James et al., 2013).

overall range of AUC as 0.65-0.72. Overall, according to the literature, about 70% accuracy and 0.75 AUC seems to be the glass-ceiling, i.e., the best available prediction, especially for future violence (Yang et al., 2010).

Predictive Tools

We have been experiencing a social transition from clinical judgment to algorithmic decision-making. Algorithms are being used in various domains, from foreign exchange markets (Chaboud, Chiquoine, Hjalmarsson, & Vega, 2014) to dating (Slater, 2013). These algorithms are generally produced within a machine learning framework. Generally considered a subfield of computer science and artificial intelligence, machine learning is concerned with designing systems that can learn from data, instead of explicitly programming them. In machine learning, the learning process is initiated with an algorithm starting with no information. As more data are fed into the system, the algorithm approximates the true value of its parameters.

Recently, there has been a surge in novel actuarial tools in criminal-justice-related outcomes, most notably recidivism. Data mining techniques are often found to be outperforming traditional methods in decreasing unexplained variance in the dependent variables (Attewell & Monaghan, 2015). Machine learning is an iterative process. It starts with an initial estimate and later improves this estimate in light of new evidence. This incremental process is known as machine learning (Attewell & Monaghan, 2015). While there has been some scholarly attention on forecasts with novel tools, from the practitioner perspective, the dominant approach seems to be logistic regression models (Yang et al., 2010).

In this section, a number of statistical approaches have been examined from the perspective of their potential in increasing predictive performance of the criminal justice system

in the U.S. The tools briefly described here are also tested against each other in later sections of this study. Below is a brief introduction to each model and by no means, it is exhaustive (for a more didactic treatment, see Hastie, Tibshirani, & Friedman, 2009).

1. Logistic Regression. Logistic regression is the generic classification tool in the social sciences and is often used for classification problems. In binary outcomes, the outcome of interest takes 0 or 1. Logistic regression is a linear combination of predictors that computes fitted probabilities (Berk, 2011). It utilizes the sigmoid function which produces probabilities (constrained between 0 and 1) for each case in the model regarding their class. Logistic regression applies maximum likelihood estimation and estimates the probability of an outcome to occur (Hosmer & Lemeshow, 1989).

Logistic regression by default sets the cut-off probability at 0.50 and assigns equal misclassification cost for the outcomes (Berk, 2012). Predicted probabilities above .50 indicate membership in one category, otherwise, they indicate membership in the other category. Equal misclassification cost indicates no constraint to the prior distributions of the variable of interest (Liu et al., 2011). If the outcome of interest has a lower probability than the absence of it, a classification will produce low false negatives compared to high false positive for the minority category (Liu et al., 2011).

Despite its common use, there are issues that need to be addressed about logistic regression. It does not effectively introduce costs, despite the fact that they are important (Berk et al., 2005). Logistic regression classification can favor the dominant category in terms of the overall proportion. Also, the nature of data (nonlinearity, etc.) may prevent from obtaining a satisfactory accuracy.

2. Random Forests. Classification trees are designed to partition the data using a given set of predictors. The decision trees are generated from the binary splitting of variables at nodes at the best split to reduce prediction error. The models utilize a bagging process which randomly selects data and cases not included in the bagged sample are used for accuracy (James et al., 2013). In other words, decision-tree-based models classify individuals associated with different outcome categories by breaking up the data into subsets of individuals with similar characteristics (Berk & Bleich, 2014). Therefore, similar cases would be grouped in the same tree branches (Breiman, Friedman, Olshen, & Stone, 1984; Breiman, 2001). The random forests method is an inductive procedure which aggregates the results from many classification trees (Berk, Sherman, Barnes, Kurtz, & Ahlman, 2009).

In a study comparing main regression approach with classification trees, Steadman et al. (2000) examined whether classification trees could be used in the development of a violence risk assessment tool. They found that the decision tree approach outperformed logistic regression. However, it is important to note that they did not use a test sample in their analysis. In another study predicting violent recidivism in probation, Stalans, Yarnold, Seng, Olson, and Repp (2004) reported that classification tree analysis provided a higher improvement in classification accuracy of violent recidivists in comparison to logistic regression. Neuilly, Zgoba, Tita, and Lee (2011) compared logistic regression with random forests and concluded that the former outperformed logistic regression. Similarly, Silver and Chow-Martin (2002) advocated using classification tree approach as an actuarial tool in the criminal justice system. There are other studies suggesting the use of classification tree approach in risk assessments (Berk et al., 2005; Berk & Bleich, 2013; Berk, Sorenson, & Barnes, 2016; Monahan et al., 2000).

3. Support Vector Machines (SVM). An SVM model forms vectors from feature space and constructs a hyperplane in multidimensional space. It uses this hyperplane to classify any new data points. That is, SVM searches for a hyperplane that divides one class from another, and it does so by maximizing the distance between the line and the data points (James et al., 2013). The more distance from a given training point results in better prediction, and these distances are called support vectors.

These support vectors are the actual data points that form hyperplanes in the input data space, and this hyperplane separate recidivists from non-recidivists (Wang, Mathieu, Ke, & Cai, 2010). The task of the hyperplane is visually described in Figure 1. The hyperplane in multidimensional space separates two classes if linear separation is possible. If that hyperplane does not exist, SVM is likely to fail to provide a satisfactory result. Though rare, one can find the applications of SVM in other crime-related research, such as crime hot spot detection (Kianmehr & Alhajj, 2008) and classification of crime scenes (Abu Hana, Freitas, Oliveira, & Bortolozzi, 2008).

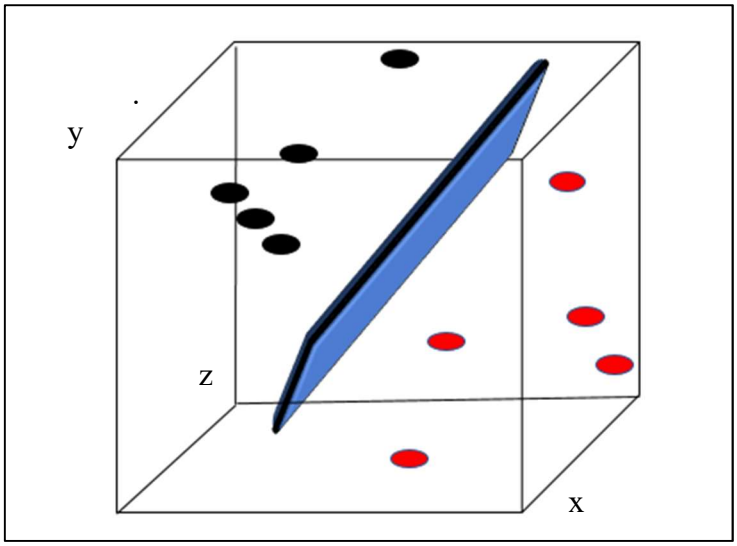

Figure 1. A hyperplane in multidimensional space.

4. XGBoost. XGBoost is an abbreviation for "Extreme Gradient Boosting", with the term "Gradient Boosting" suggested by Friedman (2001). XGBoost is a modification by Chen and Guestrin (2016) on the original ideas of Friedman (2001). It works by adding a classifier at a time, building upon the previous tree structure, whereas RF randomly and independently trains the classifier. In other words, the boosted tree model is based on a sequential logic in which each new tree utilizes information gained by the previous trees. Since it is relatively new, there have not been published criminological studies (known to the author) that have used this for risk assessment purposes. However, its close relative RF has been used quite extensively.

5. Neural Networks. Neural networks are classification models (it can be used for regression problems as well) that are designed to mimick the functionality of a human brain, specifically neurons (Rosenblatt, 1958; Warner & Misra, 1996). Neural networks are designed to resemble actual neurons in the brain, and for this reason, they are seen as information processing systems that are capable of recognizing patterns considering past experiences (Razi & Athappilly, 2005). A human brain has approximately 10^{11} neurons that are densely interconnected, and the main function of these neurons is to transmit information (for a more detailed biological treatment, see Thompson, 1985).

Neural network error reduction process is different than least squares regression's process. Multiple linear regression has a closed form of solution for the coefficients (in minimizing the loss function), but neural models utilize an iterative process (Warner & Misra, 1996). Since neural network does not have the assumption of normally distributed errors, it does not utilize confidence intervals or hypothesis testing procedures (Caulkins et al., 1996). A typical

neural network would include inputs, weights, summation function, transformation function and outputs (Razi & Athappilly, 2005).

Warner and Misra (1996) argued that neural networks can be an important asset when we do not know the functional relationship between predictors and the outcome. They utilize data when discovering these relationships, and this very fact makes neural network models data dependent, which is the reason behind the improved performance in large datasets. Regression, on the other hand, is a more appropriate tool when the sample size is small and when causal relationships are clearly defined by theory (Warner & Misra, 1996). Neural networks are known to have a high tolerance for noise and its effectiveness in dealing with nonlinearity (Razi & Athappilly, 2005).

Neural networks were previously used in predicting the presence of coronary artery disease (Kurt, Ture, & Kurum, 2008) and the recognition of Alzheimer's disease (Lehmann et al., 2007). However, a handful of studies have incorporated neural networks in predicting recidivism as well (Liu et al., 2011; Tollenaar & van der Heijden, 2013; Yang et al., 2010). Palocsay, Wang, and Brookshire (2000) argued that neural networks may provide predictive improvement for criminal recidivism outcomes in comparison to logistic regression but they also underscored that neural network performance depends heavily on the choice of the network topology, such as number of hidden layers, number of nodes in each hidden layer, and node activation functions. In their study with a sample size of 2,385 cases, Caulkins et al. (1996) failed to find an advantage of using neural network over the other methods in predicting recidivism.

6. The Search Algorithm. There can be complex combinations that might influence a social outcome. An analyst may not be aware of all possible interactions that should be factored

in the modeling process. This very problem is the exact thing Search tried to overcome. Search is a software that was designed for searching for structure by James N. Morgan and his colleagues as a result of a series of research efforts (Morgan & Sonquist, 1963; Sonquist, Baker, & Morgan, 1973). Specifically, Search splits the data sequentially on the best binary split of the best variable for reducing the unexplained variance (Morgan, 2005).

Comparative Studies

The performance of an actuarial tool is a function of specificity of predictors, well-defined criteria for outcome category, homogeneity of the targeted population, and use of an appropriate statistical model (Yang et al., 2010). Some scholars underscored the importance of the reliabilities of the information used, the kinds of measurements used, the base rate, the selection ratio used, and the representativeness of samples (Gottfredson & Moriarty, 2006). Therefore, the performance of a model can vary depending on many factors.

Recently, Tollenaar and van der Heijden (2013) compared a number of traditional and machine learning models and concluded that logistic regression and neural networks outperform other models for general recidivism. Liu, Yang, Ramsay, Li, and Coid (2011) compared logistic regression, classification and regression tree (CART), and neural networks using a sample of 1,225 cases. They reported that neural networks performed slightly better than the rival models. In his review, Harper (2005) concluded that there is no 'best' classification tool and the accuracy of prediction is dependent on features utilized in data. In a study comparing neural networks with binary logistic regression, Goss and Vozikis (2002) concluded that the prediction accuracy of neural network model was better than that of logistic regression. Similarly, Razi and Athappilly (2005) reported that both neural network and decision trees yielded better results than regression-

based approaches. In their comparative review, Berk and Bleich (2013) found that random forests perform better than logistic regression and stochastic gradient boosting. In another study, support vector machine and neural networks both outperformed logistic regression (Wang et al., 2010).

Yang et al. (2010) studied 1,353 adult men and 304 adult women prisoners for violent recidivism and 664 of man for dynamic risk factors. They compared a number of rival models, including logistic regression, discriminant analysis, classification tree, and multi-layer perceptron neural networks. They concluded that logistic regression and discriminant analysis were the most robust and least subject to over-fitting, but they also underscored the finding that their results are generally comparable.

In the following chapter, topics such as information regarding data, feature selection, and performance evaluation are discussed.

CHAPTER 3

METHODS

"*In God we trust. All others must bring data*."

— W. Edwards Deming

Data

The data used in this study were derived from the Recidivism of Prisoners Released in 1994 study developed by the U.S. Department of Justice, Bureau of Justice Statistics utilizing records of arrest and prosecution (RAP) sheets kept by state agencies and/or the FBI. This project contains information on 38,624 prisoners released from prisons in 15 states in 1994 and followed these released prisoners for three years after their release. The data have 6,427 columns consisting of 91 fields reporting information regarding before or during release from prison in 1994 (e.g., date of birth), and 64 fields that reports criminal involvement for up to 99 different times in the follow-up period repeatedly recorded for up to 99 different points in the follow-up period (for details see Langan & Levin, 2002). These data have been used in several studies, including recidivism-based investigations (see e.g., Bhati & Piquero, 2008).

The data host an impressive amount of information regarding criminal history. There is substantial detail on 99 different cycles, with each cycle documenting the most serious three offenses in each cycle as well dates regarding the arrest, conviction, confinement, total sentence length, and probation. There is information on substance abuse problems, and other information on race, ethnicity, sex, physical features (e.g., weight), and date of birth. Sources for criminal history information are state and FBI automated RAP ("Records of Arrests and Prosecutions") sheets, which includes information on arrests, adjudications, and sentences. While errors,

missing information, or differences due to jurisdiction recording policy are possible, RAP sheets are known to be the most accurate and readily available data source for repeat criminal behavior (U.S. Sentencing Commission, 2004).

Feature Selection

In addition to available data provided by BJS, with newly-created variables (details are in the following sections), there were more than 1,400 variables that could be used for the current purpose. Some of them were highly correlated, and some possibly were not so informative. If all indicator variables were to be included from all cycles, the resulting feature space would be unnecessarily complex, highly dimensional and possibly uninformative. The more complex a model gets, the more likely it overfits the data. Therefore, a strategy to reduce the number of variables to include in the models was needed. Too many variables increase the variance of a model, while too few increase the bias. Similarly, using non-informative variables increase the variance and result in overfitting (see bias-variance tradeoff, Hastie et al., 2009). An analyst prefers a model with both low bias and low variance.

In this specific dataset, inmates' prior arrest experiences were recorded indicating both time of the offense and the type of offense as the criminal history data. Most inmates had fewer than 15 prior offenses. This is evident in Figure 2. While there are inmates on the extreme end of the criminal trajectory, the majority had fewer than 15. In order to avoid unnecessary dimensionality, only 15 cycles of prior arrests were considered in feature selection process. Adding more cycles would add more dimension but offer little information on the outcome of interest. To reiterate, only 15 arrest cycles that occurred before the 1994 release were included.

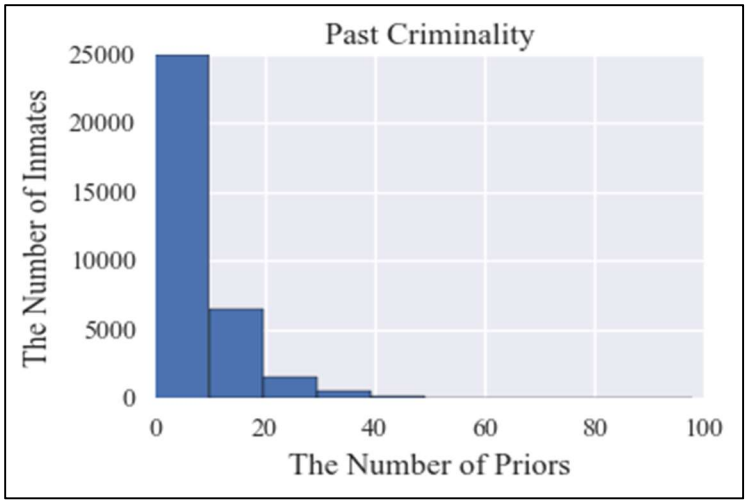

Figure 2. The number of prior arrests.

To select the most informative variables, the current study resorted to a number of dimension-reduction methods. First, highly-correlated features were removed leaving the number of variables at 397. While multicollinearity does not necessarily affect prediction (unlike inference) accuracy, extremely-correlated features do not add much to the performance of the model. Since the goal is to keep the models as simple as possible, those variables are excluded for the next stage of feature selection process, which was based on 397 variables.

Next, Least Absolute Shrinkage and Selection Operator (Lasso) was utilized to identify a subset of useful features (Tibshirani, 1996). Lasso is a technique that constrains or regularizes (shrinks) the coefficient estimates towards zero (James, Witten, Hastie, & Tibishirani, 2013, p. 214). This procedure minimizes the residual sum of squares by adding penalty equivalent to the absolute value of the magnitude of coefficients in the following manner:

$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2+\lambda\sum_{j=1}^{p}\left|\beta_j\right|$$

where n is the number of cases, p is the number of variables, βs are coefficients, and λ (also known as alpha) is the tuning or shrinking parameter, which is determined separately. When λ approaches to infinity the impact of the shrinkage penalty increases and reduces the magnitudes of the coefficients (James et al., 2013). As model complexity increases by adding more variables, the size of coefficients increases exponentially, leading to overfitting in validation process. The second term is the Lasso penalty term, which controls this mechanism by forcing some coefficients to zero values. Those variables can be seen not as useful as rest of the variables that are assigned a number different than zero.[2]

To increase the confidence in coefficient selection within the Lasso procedure, a *k*-fold cross-validation procedure is implemented. Cross-validation is a model validation technique for assessing how findings will generalize to external data (James et al., 2013). In *k-fold* cross-validation, the sample is randomly divided into *k* approximately equal subsets and one of these subsets is set aside for external validation, while *k-1* subsamples are used for training purposes (Hastie et al., 2009; Kohavi, 1995; Pedregosa et al., 2011). The entire process repeats for *k* times, resulting in a single value each time. Averaging these values provides a more robust estimate than a single train-test validation procedure. The current study employs a 10-fold cross-validation procedure in determining the optimum number of predictors in the model. A visual explanation of the *k*-fold cross-validation is shown in Figure 3. It is advantageous to use *k*-fold

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[2] Using Scikit-learn, the "LassoLarsCV" module was used to obtain coefficients. For details see Pedregosa at al. (2011).

cross-validation as it gives more accurate estimates of the test error rate than a simple train-test split validation (Hastie et al., 2009; James et al., 2013; Pedregosa et al., 2011; Raschka, 2015). To add a second layer of external validity in the process, cross-validation was conducted in training data.

Figure 3. Cross-validation. Adapted from Raschka (2015, p.176).

To avoid overfitting, the prediction performance trajectory was monitored within training sample with a focus on the mean square error on the predicted values. Doing so, the optimal value of tuning parameter was determined in a way that there is no additional gain in the performance (in terms of error) beyond that threshold. In other words, a threshold where the mean square error is the lowest was identified. Conducting this performance monitoring, the decrement in the error is shown in Figure 4, where a grid of alpha values (lambda) is plotted against the error values as a result of cross-validation procedure. Figure 4 shows the optimum tuning parameter that will suppress the uninformative coefficients toward zero, leaving useful ones with nonzero coefficients. Figure 5 documents how new coefficients emerge during this process.

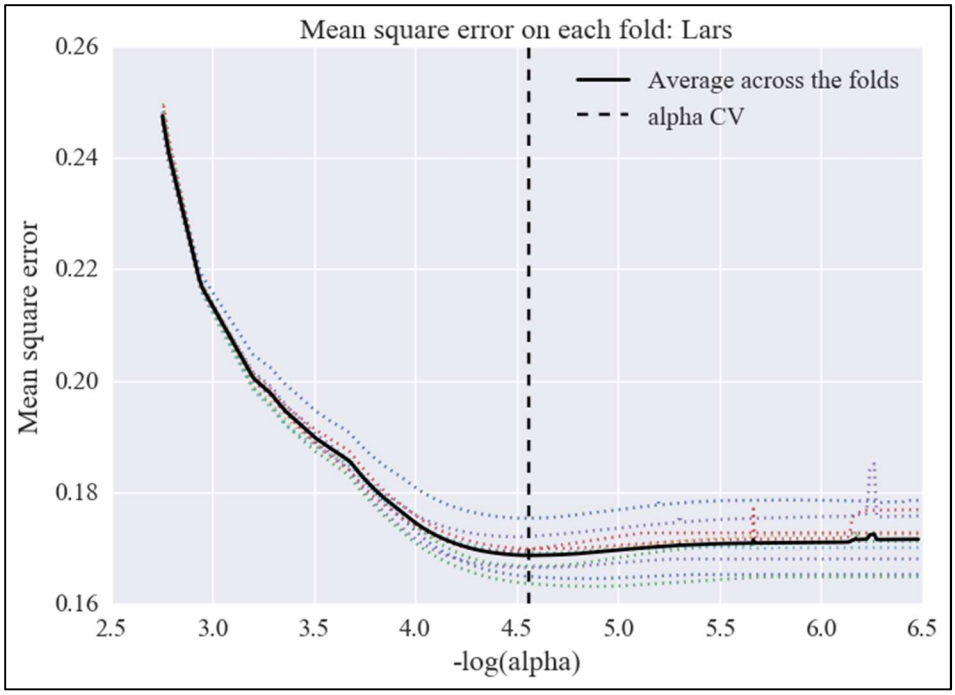

Figure 4. Selecting the optimum number of features using cross-validation.

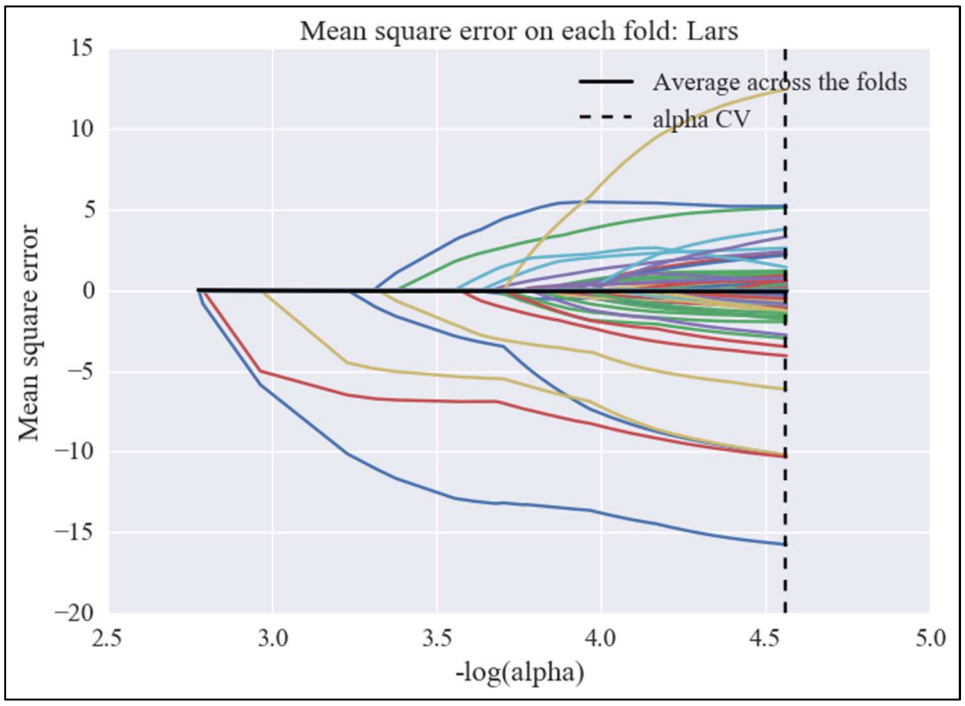

Figure 5. The emergence of relevant coefficients based on alpha values.

The above procedure yielded 150 features with non-zero coefficient values in the training data. Following the same procedure using test data, Lasso extracted 137 features that have non-zero coefficient values. Then, the same features that are identified as important by both training and test procedures were determined. This cross-checking resulted in 80 features, which were considered to have non-zero coefficients in both tests.[3] For this reason, the optimum number of the feature is selected as 80.

Lastly, it is important to note that an important variable can be associated with recidivism either positively or negatively. That is, it may decrease or increase the likelihood of recidivism depending on the coefficients' direction produced by the model. Since our goal is ultimately an increment in prediction performance, the direction of certain variables is not the actual focus in the current study. However, it is important to note that both magnitude and

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[3] In this case, train-test splits are based on 70% training and 30% test set. To investigate the veracity of the selection procedure, a different feature selection framework was implemented. Specifically, ExtraTreesClassifier, a Scikit-learn module, was performed to select the best 80 variables. Results indicated that there were 35 variables that were also suggested with the lasso procedures. Comparing relative accuracies in the sample data in the later stages, the lasso-based feature selection was deemed to be more suitable for the current study due to its higher accuracy in prediction performances of the models implemented. This comparison report is not shared in this study, but is available upon request.

direction of most of these coefficients can vary depending on the initial feature package entered into Lasso model. While some variables may be robust across various feature packages, some are highly changeable. Therefore, it is wise to focus on prediction accuracy instead of trying to make sense of individual effects provided by the model.

Variables

All variables are converted to binary form. For some originally-continuous variables, the binned categories are used. Grouping offenders surpassing a certain threshold is a practice used by the Sentencing Commission as well. For example, offenders with 20 or more priors are grouped into the last category in their report on measuring criminal history (U.S. Sentencing Commission, 2004).

1. **Criminal history related variables.** The majority of predictor variables were in this category. Details are described below.

a. Past offense types. As mentioned before, there were 99 cycles that host offenses which resulted in an arrest or absence of an arrest. In the original dataset, three variables for each cycle were used to represent three offenses committed in a given cycle. For simplicity, only the most serious offense was selected.

The data describe the first most serious arrest offense in a given cycle using a variable in which offenses were classified from 1 to 26, such as 1 = murder, 2 = voluntary manslaughter and so on. First, the most serious offenses were selected in all cycles (there were three in each cycle), then the offenses were reorganized. Specifically, the offenses were regrouped as larceny, burglary, other crimes, aggravated assault, other sexual offenses, drug possession, robbery, drug trafficking, other drug offenses, fraud, motor vehicle theft, weapon-related offenses, rape, other

property offenses, stolen property, other violent offenses, DUI, homicide, arson, manslaughter, and kidnapping for a total of 21 offense types (26 in the original, see Appendix A). Then, these offenses were reorganized as binary variables. For example, those who committed larceny were coded as 1, 0 otherwise. Therefore, for a given cycle, there were 21 binary variables indicating whether these crimes occurred in that cycle. To make criminal history representation clearer, for example, if an inmate commits robbery in all of the first five cycles, the variables representing the first cycle robbery, second cycle robbery, third cycle robbery, fourth cycle robbery, and fifth cycle robbery are all coded as 1. That inmate can commit robbery in the first and larceny in the second cycle, while not committing any crime in the remaining three cycles, thus being coded as 0 for the remaining cycles for all of the crime types. Ninety-nine cycles were re-coded in this way, but naturally the number of binary indicators decreased as the number of cycles increase.

It is important to note that not every cycle denotes a prior crime. Cycles represent the entire criminal records available to the agencies. If the release date is later than any of these cycles, then it is considered as prior. Otherwise, it denotes recidivism. The current work followed the overall practice of using a three-year period as an observation period (Bhati & Piquero, 2008; Durose et al., 2014; Langan & Levin, 2002). Offenses later than those dates are not considered technically recidivism, although they are natural acts of reoffending.

While 99 cycles were designed in a way that the most serious offense(s) was(were) included in binary form for each cycle as mentioned, only certain cycles and certain offense types were selected because they were the ones the feature selection process (previously mentioned) suggested as important variables. For example, from the first cycle, burglary, robbery, drug trafficking, other crime types, manslaughter, and rape were included, while from

the second cycle, other drug offenses, robbery, fraud, and no arrest were found to be most contributive to the predictive performance. Other related cycles can be found in Table 1.

b. Prior arrest numbers. The number of prior arrests was categorized as the number of times the inmate had been arrested before their release, including the arrest that resulted in 1994. Binary variables were constructed to indicate one prior arrest, two prior arrests, three prior arrests (reference), four prior arrests, five prior arrests, six prior arrests, 7-10 prior arrests, 11-15 prior arrests, and 16 or more prior arrests.

c. Age at first arrest. This variable indicates the age at which an inmate was first arrested. Among the binary-coded variables for the ages 0-17, 18-24, 25-29, 30-39, and 40 plus, only the latter was suggested by the Lasso.

d. Prior Cycle Outcome. Each arrest cycle had certain outcomes. If an outcome for a given cycle is jail or prison, that outcome for that cycle was coded as 1, otherwise 0. Based on the feature selection, among the first 15 cycles, 11th and 12th incarceration results were suggested by the Lasso procedure.

e. Out-of-state arrest. If the inmate was ever arrested out of state, this variable was coded 1, otherwise 0.

f. Prior Sentence. This binary coded variable indicates whether, prior to the sentence that got the inmate into the study, he/she had previously received a prison sentence for another offense. This variable is different than individual cycle outcomes as jail or prison since it indicates an overall incarceration experience, rather than cycle-based incarceration.

Table 1. *Descriptive information.*

Variables	N	Mean	Std	Min	Max
One Prior	29323	0.13	0.34	0	1
Two Priors	29323	0.11	0.32	0	1
Four Priors	29323	0.09	0.28	0	1
Five Priors	29323	0.08	0.27	0	1
Six Priors	29323	0.07	0.26	0	1
7-10 Priors	29323	0.19	0.39	0	1
11-15 Priors	29323	0.12	0.32	0	1
16-Plus Priors	29323	0.12	0.32	0	1
Age 14_17	29323	0	0.04	0	1
Age 18_24	29323	0.21	0.41	0	1
Age 25_29	29323	0.21	0.41	0	1
Age 35_39	29323	0.16	0.37	0	1
Age 40_44	29323	0.1	0.3	0	1
Age 45_Plus	29323	0.11	0.31	0	1
Female	29323	0.07	0.25	0	1
Black	29323	0.41	0.49	0	1
Hispanic	29323	0.16	0.37	0	1
Robbery_Offense	29323	0.07	0.25	0	1
Rape	29323	0.08	0.27	0	1
Traffic_Offense	29323	0.04	0.18	0	1
Murder	29323	0.02	0.13	0	1
Manslaughter	29323	0.02	0.13	0	1
Motor Vehicle Theft-Larceny	29323	0.08	0.28	0	1
Cycle-1 Crime Burglary	29323	0.01	0.12	0	1
Cycle-1 Crime Robbery	29323	0.06	0.24	0	1
Cycle-1 Crime DrugTrf	29323	0.05	0.22	0	1
Cycle-1 Crime Other	29323	0.13	0.33	0	1
Cycle-1 Crime Manslaughter	29323	0	0.05	0	1
Cycle-1 Crime Rape	29323	0	0.05	0	1
Cycle-2 Crime OtherDrug	29323	0.03	0.18	0	1
Cycle-2 Crime Robbery	29323	0.05	0.22	0	1
Cycle-2 Crime Fraud	29323	0.03	0.18	0	1
Cycle-2 Crime No Arrest	29323	0.08	0.27	0	1
Cycle-3 Crime Burglary	29323	0.08	0.28	0	1
Cycle-3 Crime Stolen Property	29323	0.02	0.12	0	1
Cycle-3 Crime Manslaught	29323	0	0.03	0	1
Cycle-3 Crime No Arrest	29323	0.15	0.36	0	1

Variables	N	Mean	Std	Min	Max
Cycle-3 Crime Other Sexual Offenses	29323	0.04	0.19	0	1
Cycle-3 Crime Rape	29323	0.02	0.14	0	1
Cycle-4 Crime Other Violence	29323	0.02	0.13	0	1
Cycle-4 Crime Robbery	29323	0.04	0.19	0	1
Cycle-4 Crime Stolen Property	29323	0.01	0.11	0	1
Cycle-4 Crime No Arrest	29323	0.22	0.42	0	1
Cycle-4 Crime Other Sexual Offenses	29323	0.03	0.16	0	1
Cycle-4 Crime Rape	29323	0.02	0.12	0	1
Cycle-5 Crime Fraud	29323	0.02	0.16	0	1
Cycle-5 Crime Rape	29323	0.01	0.12	0	1
Cycle-5 Crime No Arrest	29323	0.29	0.45	0	1
Cycle-6 Crime Arson	29323	0	0.03	0	1
Cycle-6 Crime Agg. Assault	29323	0.04	0.2	0	1
Cycle-6 Crime OtherDrug	29323	0.02	0.14	0	1
Cycle-6 Crime Other Sexual Offenses	29323	0.02	0.13	0	1
Cycle-7 Crime Larceny	29323	0.06	0.23	0	1
Cycle-7 Crime Drug Possession	29323	0.03	0.17	0	1
Cycle-7 Crime Other Violence	29323	0.01	0.11	0	1
Cycle-7 Crime Motor Vehicle Theft	29323	0.01	0.11	0	1
Cycle-7 Crime Other Sexual Offenses	29323	0.01	0.11	0	1
Cycle-8 Crime Other Property	29323	0.01	0.1	0	1
Cycle-9 Crime Kidnap	29323	0	0.03	0	1
Cycle-9 Crime Other Property	29323	0.01	0.09	0	1
Cycle-9 Crime Other Violence	29323	0.01	0.09	0	1
Cycle-10 Crime Arson	29323	0	0.02	0	1
Cycle-10 Crime Larceny	29323	0.04	0.19	0	1
Cycle-10 Crime Drug Trafficking	29323	0.02	0.13	0	1
Cycle-11 Crime Weapon-Related	29323	0.01	0.08	0	1
Cycle-11 Crime Motor Vehicle Theft	29323	0.01	0.08	0	1
Cycle-12 Crime Other Sexual Offenses	29323	0	0.06	0	1
Cycle-12 Crime Arson	29323	0	0.02	0	1
Cycle-13 Crime Larceny	29323	0.03	0.17	0	1
Cycle-13 Crime Motor Vehicle Theft	29323	0.01	0.08	0	1
Age of First Arrest 40_Plus	29323	0.05	0.21	0	1
Expiration of Sententence	29323	0.1	0.29	0	1
Probation-Release-Shock Probation	29323	0.04	0.19	0	1
Parole Brd. Decision No Minimum	29323	0.26	0.44	0	1
11th Prior Incarceration	29323	0.09	0.28	0	1

Variables	N	Mean	Std	Min	Max
12th Prior Incarceration	29323	0.07	0.26	0	1
Time Served 7_12 Months	29323	0.18	0.39	0	1
Time Served 37_60 Months	29323	0.11	0.32	0	1
Prior_Sentence	29323	0.26	0.44	0	1
Out of State Arrest	29323	0.21	0.41	0	1
Rearrest	29323	0.58	0.49	0	1

Note: Variables not included in Table 1 served as categorical references, such as three priors, age 30-35, or Non-Black and Non-Hispanics.

2. **Demographic-related variables.** Sex, race and age factors are included as demographic features.

a. Sex. Sex is coded as 1 and 0 for females and males, respectively.

b. Race. African Americans and Hispanics were binary coded as 1 indicating that they belong to these categories, 0 otherwise. Those who are not in either category were represented in the reference category.

c. Age category. Seven binary age categories are created: 14-17, 18-24, 25-29, 30-34, 35-39, 40-44, and 45 plus. Age category for 30-34 served as the reference category.

3. **Sentence-related variables.** The sentence length refers to the sentence for which the inmate was in prison in 1994. It is categorized in binary forms, including 1-6 months, 7-12 months, 13-18 months, 19-24 months, 25-30 months, 31-36 months, 37-60 months, and 61 or more months. The variables for 7-12 and 37-60 months were the only ones used in the models as suggested by the Lasso procedure.

4. **Current offense related variables (that resulted in incarceration before 1994 release)**. Offense types were combined based on similarity and intensity of the criminal acts. Based on the information provided by the BJS, crimes are categorized in the following manner:

other sexual crimes, drug trafficking, burglary, motor vehicle theft or larceny, other, robbery, rape, drug possession, aggravated assault, traffic-related, other violent crimes, fraud, murder, manslaughter, and kidnap. There were some regroupings for the sake of simplicity and practicality. For example, the murder of a police officer, murder, attempted murder, conspiracy murder, and unspecified murder was combined to form a general "murder" category. Homicide typology is important for recidivism. Prior studies found a significant difference between the type of homicide committed and whether or not the offenders recidivated (Roberts et al., 2007). The feature selection process indicated that traffic-related offenses, rape, robbery, murder, manslaughter, and motor vehicle theft or larceny were the most informative ones and therefore if the inmate was serving for any of these crimes, they are coded as 1, otherwise 0. The detailed account of forming offense types can be found in Appendix B.

5. **Release type**. The type of release from correctional facilities in 1994 was described in the study. The release types included in the models were the expiration of sentence, probation release-shock probation, and parole board decision with no minimum. All other categories used as the reference category. All of these variables are coded as 1 if the inmate was released based on the type described in the study, as 0 otherwise.

Analytic Procedure

The current study attempts to conduct supervised learning. In supervised learning, the analyst knows the actual values to be found in the training set to train the algorithms. The algorithm iteratively adjusts the model weights, and after finding the optimum values, learning structure is tested on test data. Finally, model performance is evaluated using a predefined accuracy metrics.

Specifically, a number of rival models—most of them were built using Scikit-learn, a Python computation environment, see Pedregosa et al. (2011)— were compared in terms prediction performance in the current study. The Scikit-learn platform was built by a collective effort of several contributors and it is open for modification suggestions coming from the community (Pedregosa et al., 2011).

The rival models were determined as logistic regression, random forests, support vector machines, XGBoost, neural network, and Search algorithm. Before model estimation, missing cases were discarded. This does not pose a challenge as the main concern was not inference. Imputing cases, on the other hand, can increase or have a null effect on prediction accuracy. This issue was left for a future inquiry using the same data.

Another important analytic decision was using test data to avoid overfitting. It is a common practice to hold out part of the available data as a test set to evaluate the performance of a predictive function. This hold-out sample was created randomly, and all parameter estimates and model performance metrics reported later in the study were obtained using the test set, which was 30% of the available data at hand.

Each model has its own fine-tuning procedure, that is, there are hyperparameters and optimization choices the analyst should decide to include in the models. These preferences can affect accuracy considerably. For example, in the case of a random forest, how deep a tree structure one would prefer? At which point one would like to stop training? If one does not constrain this, after a certain point, the algorithm starts modeling the noise, which causes overfitting problem in a validation set. These are essential things an analyst should consider,

most of the time, by trial and error if there is no intuition or prior experience on the problem.[4]

Hyperparameters are parameters that are passed as arguments to the constructor of the estimator classes (Pedregosa et al., 2011). The grid search (GridSearchCV in Scikit-learn) exhaustively generates candidates from a grid of parameters specified by the analyst (Pedregosa et al., 2011). This procedure exhaustively considers all parameter combinations defined by the user. In other words, a combination of potential hyperparameters is used to find which combination yields the best prediction accuracy.

For example, let us assume the parameter "n_estimators" (number of trees) is defined as a vector of possible values 10, 50, and 100. Similarly, the parameter "max_depth" (the depth of the forest) is defined as 3, 5, and 10 by the analyst. The grid search searches various combinations (such as n_estimators = 10 and max_depth = 3 or n_estimators = 10, and max_depth = 5 and so on) that yields the best accuracy. Obviously, this is a computationally and temporally expensive process, but it can provide rewarding outcome by pointing out the best combination.

To increase our confidence in the hyperparameter selection, a 10-fold cross-validation procedure was used within this grid search procedure. The same logic that is explained in Figure 3 was applied in finding the optimum parameters. In other words, these parameters were cross-

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[4] For example, potential parameter choices in random forests (in Scikit-learn) are as follows: RandomForestClassifier(n_estimators=100, criterion='gini', max_depth=5, min_samples_split=2, min_samples_leaf=1, min_weight_fraction_leaf=0.0, max_features='auto', max_leaf_nodes=No) ne, min_impurity_split=1e07, bootstrap=True, oob_score=False, n_jobs=1, random_state=None, verbose=0, warm_start=False, class_weight=None).

validated. The advantage of this procedure is that it provides highly-reliable estimates regarding the values to choose from a list of options for a given hyperparameter. The grid search procedure was used for all rival models to get the most of these models. Specifically, logistic regression, random forests, support vector machines, XGBoost, and neural networks were processed with grid search procedure to obtain the best hyperparameters.[5] The following section reports models with the resulting hyperparameters that were suggested by the grid search. A random number (3) was chosen as a seed to make the replication possible in all models.

1. **Logistic regression.** L1 regularization was chosen. This helps avoid overfitting the model. For details about L1 regularization, see Hastie et al. (2009).

2. **Random Forests.** Unlike logistic regression, there are plenty of parameters to choose in random forests. Specifically, the number of features to consider when looking for the best split was chosen as "auto", the minimum number of samples required to split an internal node was chosen as eight, the number of trees in the forest was chosen as 125, and the maximum depth of the tree was chosen as 10.[6]

3. **Support Vector Machines.** The hyperparameters chosen to fine tune were kernel choice and gamma. The former specifies the kernel type to be used in the algorithm among possible choices such as 'linear', 'poly', 'rbf', 'sigmoid', or 'precomputed. Grid search yielded "rbf" as the best kernel choice. Rbf is a kernel that is in the form of a radial basis function (a

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[5] Unfortunately, grid search was not possible for the Search algorithm.

[6] Parameters: max_features= 'auto', min_samples_split= 8, n_estimators = 125, max_depth=10.

Gaussian function). The latter, gamma is coefficient for the kernel, and it was identified as gamma= 0.05.[7]

4. XGBoost. As for XGBoost, the number of boosted trees to fit was selected as 300, the maximum depth of a tree was identified as five, and learning rate—adjusts the learning speed—was identified as 0.05. The learning speed in machine learning literature, in general, indicates the amount of corrections being made in the parameters in the case of false prediction.

5. Neural Networks. The goal of the neural network is to detect patterns in the same way that the human brain would. A neural network project operates on multiple neural units and connections. Each neural unit is connected with other units. These links can be enforcing or inhibitory in their effect on the activation state of connected neural units.

Following Warner and Misra (1996), the functionality of a neuron can be explained as follows: The neuron receives a weighted sum of inputs from connected units and yields a value of one (firing) if this sum is greater than a threshold. If the sum is less than the threshold, the neuron yields s a zero value (Warner & Misra, 1996). Mathematically;

$$y_i = \theta\left(\sum_j w_{ij}\, x_j - \mu_i\right)$$

where y_i is the output of neuron i, w_{ij} is the weight from neuron j to neuron i (also called synaptic weight). It is interpreted as the strength of the connection from the jth unit to the ith unit.

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[7] Parameters: kernel = rbf and gamma = 0.05.

x_j is the output of neuron *j,*

μ_i is the threshold for neuron i,

θ is the activation function, defined as:

$$\theta(\text{net input})=\begin{cases} 1, & \text{if net inputs} \geq 0 \\ 0, & \text{Otherwise} \end{cases}$$

Neural networks are powerful tools, but they come with a cost as they are hard to train. Decisions regarding model building include but are not limited to the number of neurons, the number of layers, activation, optimization, batch size, etc. The way hidden layers are constructed can heavily affect the generalization capability of the network as omitting required hidden layers cause underfitting or excessive use may result in overfitting (Harper, 2005).

The neural network used in this study was built using Keras, which is a high-level neural networks library, written in Python (Chollet, 2015). Keras allows specification of a model to organize layers. This practical convenience is provided by the sequential model, a linear stack of layers (Chollet, 2015). For a more exhaustive introduction on Keras, see Chollet (2015).

To fine-tune neural networks, an iterative method was used in which sequential numbers of neuron and layers were tested as well as necessary optimization choices, just as the way it was done for random forests or XGBoost. Of course, fine-tuning deep learning models is a very challenging process that includes many optimization choices. To overcome these problems, Keras models were wrapped to use in Scikit-learn and grid search (i.e., KerasClassifier, see Brownlee, 2016).

The resulting neural structure was a three-layer topology with 10 neurons in the first hidden layer, 40 neurons in the second, and finally 4 neurons in the last hidden layer. In the data

science literature, neural networks are called “deep learning” when there is more than one hidden layer, similar to the neural structure built in this study. As the number of layers and neurons increase, the training sample accuracy increases, but it becomes less likely to fit equally well the external data. Therefore, model building (as well as feature selection beforehand) becomes more art than science, especially in deep learning specification. There are many parameters one can adjust in building a deep learning model.[8]

In Figure 6, one can see the general topology of the neural network used. In this topology, there are 80 inputs or features, all of which are fed into 10 neurons in the first hidden layer. Then, the output from each 10 neurons is fed into 40 neurons in the second hidden layer. Next, outputs of these 40 neurons are transferred to the last hidden layer, which has only four neurons. Last, outputs of these four neurons are transferred to the output neuron, which fires based on its activation function (or not fire—no recidivism).

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[8] First layer: 40 neurons, input_dim=80, init='uniform', and activation='relu.’

Second hidden layer: 40 neurons, init='uniform', and activation='relu.'

Third hidden layer: 4 neurons, init='uniform', and activation='relu.'

Output layer: one neuron, init='uniform', and activation='sigmoid.'

Model compilation: loss='binary_crossentropy', optimizer='adadelta', and metrics='accuracy.'

Fitting the model: nb_epoch=40 and batch_size=40.

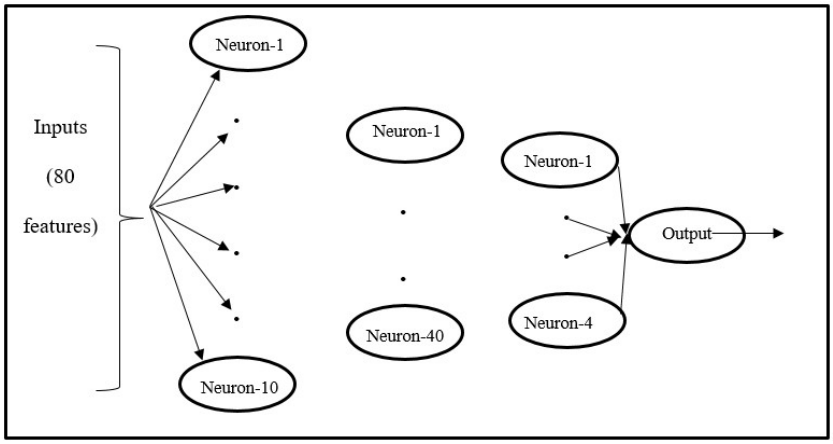

Figure 6. A three-layer deep learning structure.

Neural networks learn through a mechanism called backpropagation. The use of backpropagation allows finding a solution to a problem that is not linearly separable (Warner & Misra, 1996). Backpropagation is what neural networks use to update weights. This is an iterative process as well, which continues until an optimum solution is found. When the model makes a wrong prediction, it updates the weights that are attached to units. If the prediction is correct, the weights do not change. For more detailed treatment on backpropagation, see Samarasinghe (2007). A visual description of how backpropagation operates is explained in Figure 7.

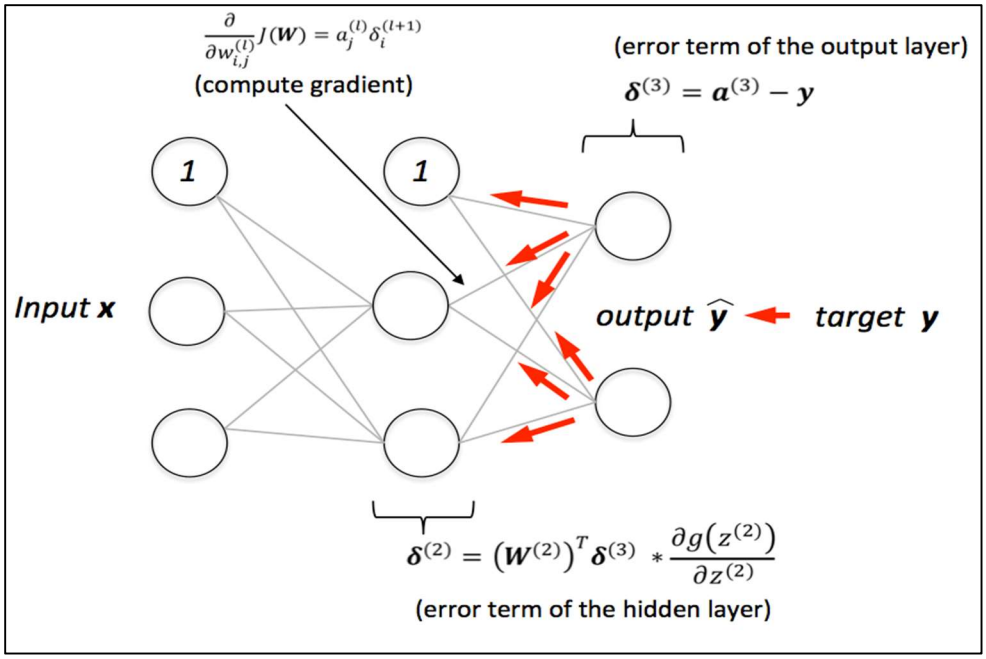

Figure 7. Backpropagation mechanism. Adapted from Raschka (2015, p. 371).

6. Search. As for Search algorithm, no fine-tuning effort was done. Simply, the input features were fed into the Search program, and then the program was used to look for structure in the training set. Next, similar to other models, the resulting structure was tested using the test data. The next chapter provides details about the results of all aforementioned models.

CHAPTER 4

RESULTS

"*Machines take me by surprise with great frequency.*"

— Alan Turing

Until now, the feature selection process revealed most contributing variables that can be used in the machine learning models. Then, a number of models—logistic regression, random forests, support vector machines, XGBoost, neural network, and Search algorithm—were selected for comparison to see which can be used to increase the predictive performance in the criminal justice system, within the context of recidivism. The following section presents the outcomes obtained by these models.

Performance Metrics

There are four possible outcomes after any prediction. The predicted case can be true positive, true negative, false positive, and false negative. False positive and false negatives are not desired outcomes because the model erroneously predicts a non-recidivist as recidivist (false positive) and a recidivist as non-recidivist (false negative) in the criminal justice setting. False positive predictions indicate a waste of time and resources on someone who will not recidivate, whereas false negative predictions indicate crimes that could have been prevented. In the case of violent offending, it is safe to assume that false negatives are one of the main kinds of cost that the criminal justice system intends to avoid in the first place. Alternatively, for other decision-makers, a false positive would represent an attack on due process.

There are several ways to judge the performance of a predictive model. Rates of false positives and false negatives, sensitivity, precision, specificity, F1-score, and the area under the

curve (AUC) are the most used performance measures. Below is a brief definition of these methods.

$$\text{False Positive Rates} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$$

$$\text{False Negative Rates} = \frac{\text{False Negatives}}{\text{False Negatives} + \text{True Positives}}$$

A more nuanced elaboration on these measures is *recall* and *precision.* High recall indicates that the model does a good job in detecting recidivists, but at the cost of labeling non-recidivists as a recidivist. This is similar to using a large net for catching fish (Watson, 2016). The large net will surely catch more fish along with other things we do not want to catch. A high precision model, on the other hand, is not as ambitious as detecting the outcome of interest as a high recall model because it does a conservative job to detect only the outcome of interest. This model would detect fewer recidivists, but it also produces fewer false positives.

$$\text{Sensitivity (recall)} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

- If a person is a recidivist, how often will the model identify him as recidivist? Sensitivity, in this case, refers to the percentage of recidivists who are correctly identified as recidivists. In other words, if the model is highly sensitive, and the test result is non-recidivist, one's confidence in the result can be substantially high due to high sensitivity.

$$\text{Precision (positive predictive value)} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

- Precision is about relevancy. In this case, the relevancy indicates actual recidivism, and precision is the probability that a positively-labeled case is relevant.

$$\text{Specificity (true negative rate)} = \frac{\text{True Negatives}}{\text{True Negatives + False Positives}}$$

- If a person is not a recidivist, how often will the model identify him as non-recidivist?

Specificity refers to the percentage of non-recidivist who are correctly identified as non-recidivist. If the model has a high specificity and the prediction is a recidivist, one can be highly confident that the inmate is a recidivist.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision+ Recall}}$$

- The F1 score combines precision and recall by taking the harmonic mean of precision and recall. It can be seen as a middle ground between precision and recall.

According to Table 2, logistic regression correctly identifies 4541 people as recidivist and 2178 people as non-recidivist. The number of false negatives for logistic regression is 590, while the number of false positives is 1488. Logistic regression obtained about 0.76 test accuracy, which reflects an improvement over the glass-ceiling accuracy score documented in prior studies. Another important finding is that logistic regression is highly stable. There is no sign of overfitting (0.762 versus 0.762).

Table 2. Classification by logistic regression.

	Predicted-No Recidivism	Predicted-Recidivism	Total
Actual Outcome-No Recidivism	2178	1488	3666
Actual Outcome-Recidivism	590	4541	5131
Total	2768	6029	8797

Training vs Test Accuracy: 0.762 and 0.762 respectively.

When the classification outcome for random forests is examined in Table 3, one can see that there is an increased predictive accuracy in true positives in comparison to logistic

regression. 4747 cases are correctly identified as recidivists, while 1916 cases are correctly labeled as non-recidivists. While random forests provided an elevated prediction accuracy in true positives, it produced less true negatives. The discrepancy between training and test accuracy does not show a serious sign of overfitting (0.767 versus 0.757).

Table 3. Classification by random forests.

	Predicted-No Recidivism	Predicted-Recidivism	Total
Actual Outcome-No Recidivism	1916	1750	3666
Actual Outcome-Recidivism	384	4747	5131
Total	2300	6497	8797

Training vs Test Accuracy: 0.767 and 0.757 respectively.

As for support vector machines' performance outcome in Table 4, 4852 cases are correctly labeled as recidivists and 1893 cases as non-recidivists. There is no sign of overfitting (0.770 versus 0.766), and it is possible to argue that support vector machines provided better, though slightly, classification accuracy than logistic regression and random forests.

Table 4. Classification results for support vector machines.

	Predicted-No Recidivism	Predicted-Recidivism	Total
Actual Outcome-No Recidivism	1893	1773	3666
Actual Outcome-Recidivism	279	4852	5131
Total	2172	6625	8797

Training vs Test Accuracy: 0.770 and 0.766 respectively.

According to Table 5, XGBoost correctly identifies 4783 cases as recidivists and 2064 cases as non-recidivists. XGBoost did a superior job in terms of providing a better overall classification accuracy. The discrepancy between training and test sets is acceptable and not worrisome. In the case of overfitting, though, there are a number of remedies that can be included in these kinds of model, such as early stopping, but so far this is not needed. XGBoost

attaches a certain importance to features, and this can be seen visually in Figure 8. According to Figure 8, XGBoost attached more importance on release type (parole board-no minimum), race, overall incarceration, and out-of-state arrests than the remaining variables. Whether someone committed arson in his tenth cycle seemed to be least important, according to the XGBoost.

Table 5. Classification results for XGBoost.

	Predicted-No Recidivism	Predicted-Recidivism	Total
Actual Outcome-No Recidivism	2064	1602	3666
Actual Outcome-Recidivism	348	4783	5131
Total	2412	6385	8797

Training vs Test Accuracy: 0.794 and 0.778 respectively.

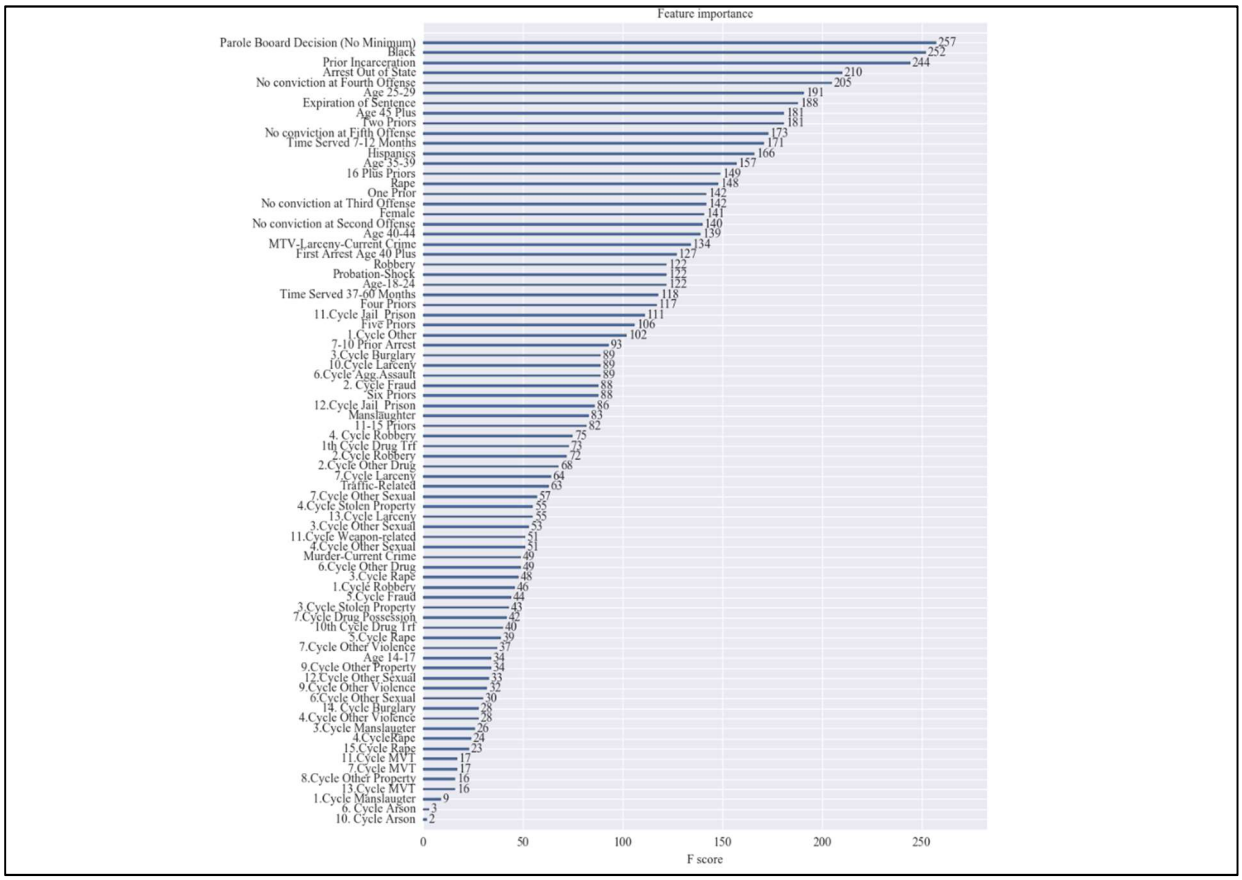

Figure 8. Feature importance according to XGBoost.

According to Table 6, another high-performance model is neural networks. In this model, 4773 cases are correctly labeled as recidivists and 2050 cases as non-recidivists. Neural network output is very comparable to what XGBoost produced, though there is less difference between training and test sets. It is important to note that if neural networks cannot be constructed appropriately, they easily tend to overfit and increase the variance of the model substantially.

Table 6. Classification results for neural networks.

	Predicted-No Recidivism	Predicted-Recidivism	Total
Actual Outcome-No Recidivism	2050	1616	3666
Actual Outcome-Recidivism	358	4773	5131
Total	2408	6389	8797

Training vs Test Accuracy: 0.784 and 0.775 respectively.

Table 7 presents the prediction results for the Search algorithm. This model correctly identified 1922 cases as non-recidivist and 4822 cases as recidivists. The performance of the Search is comparable with the other models in comparison. A more detailed output produced by Search can be found in Appendix C. Also, a visually informative description of how Search partitions the variables can be found in Appendix D.

Table 7. Classification results for the Search algorithm.

	Predicted-No Recidivism	Predicted-Recidivism	Total
Actual Outcome-No Recidivism	1922	1744	3666
Actual Outcome-Recidivism	309	4822	5131
Total	2231	6566	8797

Training vs Test Accuracy: 0.766 and 0.766 respectively.

Performance Assessment

The most important concern in any given predictive study is whether the model provides better prediction than random chance (e.g., guessing that everybody recidivates will be correct

almost half of the time). There is also another performance measure that is more conservative than the better-than-chance measure, the null accuracy. This refers to always guessing the majority class (e.g., if there is 0.70 percent recidivism rate, one can be correct 70 percent of the time by always guessing that the inmate will recidivate).

Therefore, any model that claims to make forecasts should be better at predicting outcomes better than random chance. As for the null accuracy, while it is also desirable to beat the null accuracy, the best accuracy may not be an indicator of the most useful model due to relative cost sensitiveness of the stakeholder. Related to this issue, a ROC curve that judges the performance of a model using a graph that plots the true positive rates (TPR) against the false positive rates (FPR). Using the area under the curve as the predictive power of the tool will satisfy the aforementioned two concerns regarding prediction performance (Bradley, 1997).

There are multiple performance measures varying on the need of the analysis. For example, if a department wants to detect future recidivists of sex offenses regardless of its cost, a model that has a high recall rate might be preferable. If it prefers some sort of balance, then an F1 score could be a viable alternative as a measuring rod.

In order to provide a broader picture, Table 8 provides a summary table including all necessary information regarding evaluating model performances. This table includes raw accuracy scores (how many cases were correctly labeled?), AUC, sensitivity (known as hit rate), specificity, false prediction rates, and F1-Score. All of these measures can be used for evaluating a model in question. If one desires a model that predicts positives as much as possible regardless of the number of false positives, then recall can be a good evaluation metric. Similarly, a high precision ensures that fewer cases are erroneously labeled as the outcome of interest.

Table 8. Comparison of the models using various performance criteria.

Metric	Logistic Regression	Random Forests	Support Vector Machines	XGBoost	Neural Networks	Search
Training Accuracy	0.762	0.767	0.770	0.794	0.784	0.766
Test Accuracy	0.762	0.757	0.766	**0.778**	0.775	0.766
AUC	0.813	0.813	0.806	**0.824**	0.822	0.793
Sensitivity	0.885	0.925	**0.946**	0.932	0.930	0.940
Precision	**0.753**	0.731	0.732	0.749	0.747	0.734
Specificity	**0.594**	0.523	0.516	0.563	0.559	0.524
False Positive Rate	**0.406**	0.477	0.484	0.437	0.441	0.476
False Negative Rate	0.115	0.075	**0.054**	0.068	0.070	0.060
F1-score	0.813	0.816	0.825	**0.830**	0.828	0.824

According to Table 8, one can see that all models provided somewhat comparable outcomes with slight differences. The best values are shown in bold-face across the rival models. Specifically, XGBoost had the highest test accuracy (0.778), AUC value (0.824), and F1-score (0.830), while support vector machines had the lowest false negative rate (0.054). As far as the hit rate (sensitivity) is concerned, support vector machines are the clear winner (0.946), followed by Search algorithm (0.940). If one desires a balanced predictive power that is good for detecting positive cases, but also careful in not erroneously labeling cases as positive, F1-score can be used as a general metric of the predictive performances. Doing so, XGBoost provided the best performance (0.830), while neural networks immediately followed XGBoost at 0.828. Also, despite their different working mechanisms, XGBoost and neural networks yielded almost identical results. Logistic regression provided the smallest F1-score among all models.

Asymmetric Cost

In previous sections, the importance of differential cost evaluation was discussed at length. Briefly, a parole board may want to have a low False Negative Rate (rather than a low False Positive Rate) to avoid future crimes, especially violent ones (Berk & Bleich, 2014). To include this sensitivity, a range of different cost ratios will be tested.

When the rates of false negatives and false positives are about equal, there is an equal cost for making the wrong prediction in either direction. A different threshold translates into a cost of falsely forecasting a recidivist is differently than the cost of falsely forecasting a non-recidivist (Berk, 2011). While there is no hard and fast rule about where the threshold should be, it can be easily adjusted according to a practitioner's goals, needs, and priorities.

Due to its high performance, XGBoost will be used in this section with adjusted probability thresholds that assign a case to recidivist class. If the threshold is set as say 0.25, then we are explicitly making the model sensitive to missing actual recidivists at the cost of denying parole to some. The p-values shown below indicate the thresholds after which a case is assigned to recidivism.

- $p > 0.2$ False Negatives are about 109 times costlier than false positives.
- $p > 0.3$ False Negatives are about 48 times costlier than false positives.
- $p > 0.4$ False Negatives are about 18 times costlier than false positives.
- $p > 0.6$ False Negatives are 2 times costlier than false positives.
- $p = 0.665$ False Negatives and False Positives have the same cost.
- $p > 0.8$ False Negatives are 0.14 times costlier than false positives (missing a recidivist is

much less important than keeping a non-recidivist in prison).[9] These thresholds were arbitrarily chosen, therefore do not indicate any optimum level.

A sensitivity in recall is concerned with the variation of the number different false negatives that a model potentially produces. This is visually shown in Figure 9. One can easily see that if the classifier's threshold is dragged from the 0.5 to the 0.4 level, one can gain an increase in hit rate with increase in false positives. Related classification results for the 0.4 threshold can be found in Table 9. As can be seen, the needs of an agency can be adjusted. This adjustment can reflect different needs and sensitivity of criminal justice agencies for various situations. If the department wants to detect say, most future sex offenders, technically this can mostly be realized by the inclusion of differential costs into decision-making process (this will happen at the cost of denying parole to many).

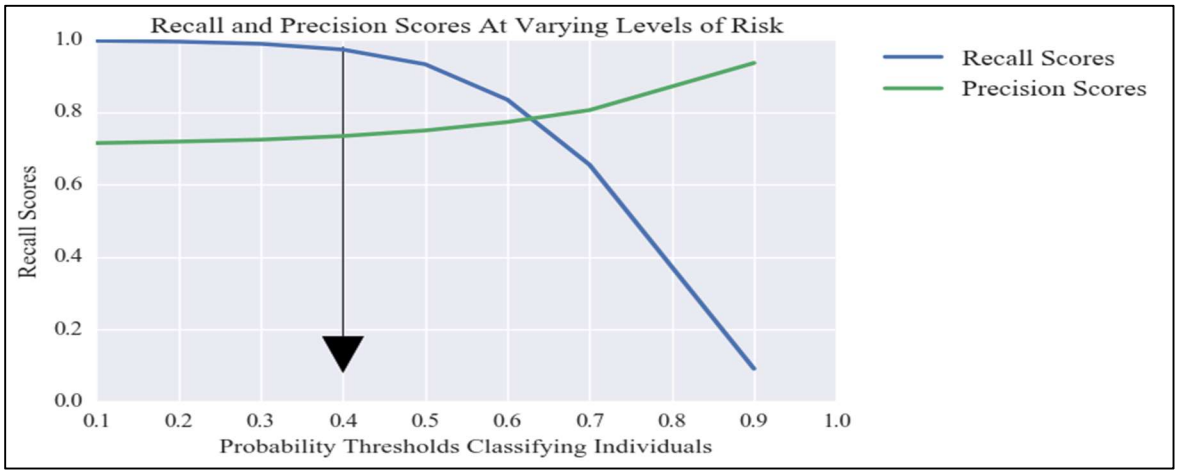

Figure 9. Differential costs of decision-making.

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[9] For the $p < .2$, 109 was obtained dividing the False Positive Rate of 0.546 by the False Negative Rate of 0.005. Others were similarly calculated.

Table 9. Classification results for the cost-sensitive XGBoost.

	Predicted-No Recidivism	Predicted-Recidivism	Total
Actual Outcome-No Recidivism	1855	1811	3666
Actual Outcome-Recidivism	140	4991	5131
Total	1995	6802	8797

AUC: 0.824, Sensitivity: 0.973, Precision: 0.734, Specificity: 0.506, FNR: 0.027, FPR: 0.494.

We can see that criminal justice stakeholders can decide on the relative importance of making an erroneous decision according to their priorities. This decision can be influenced by the needs of the community being served, budgetary considerations, or other criteria. As can be seen in Figure 9, when one increases the threshold toward detecting more recidivists, the number of detected recidivists increases (lower false negatives), but this comes at the cost of the increased number of erroneously-labeled non-recidivists (higher false positives). For example, a policy that is 18 times more aggressive on recidivist-detection can miss only a handful of cases who actually recidivate (140 in Table 9), but also deny parole or probation to about 1811 cases although they would not recidivate.

CHAPTER 5

CONCLUSION

"*When the facts change, I change my mind. What do you do, sir?*"

— John Maynard Keynes

The current study sought ways to increase the predictive power that is available to the U.S. criminal justice system within the specific context of inmate recidivism. Using a large volume of information from the BJS, this study explored a number of statistical models in terms of their performance at predicting recidivism. Also in line with real world problems, explicitly recognizing differential costs of making false predictions, the current work presented policy-wise decision-making solutions in an effort to build upon the prior literature. The risk is defined as the likelihood of recidivism, and risk factors increase—whereas protective factors decrease—this likelihood. With this atheoretical approach, this study does not claim to identify any causal mechanisms, instead arguing in favor of the use of more nuanced algorithms that can increase the classification accuracy and reduce classification costs.

Incarceration is costly. Overall, the criminal justice system has cost taxpayers over $270 billion with a huge increase in the past two decades. In 2013, 11 states spent more on incarceration than on higher education (The Council of Economic Advisors, 2016, p.5). There are other types of unmeasured costs, or collateral consequences, for individuals and for society. These indirect costs can include difficulty in finding employment, increase poverty within the family, decrease the likelihood of marriage, increase the likelihood of divorce, and other within-family problems, such as mental health problems or school dropout (The Council of Economic Advisors, 2016).

One way to tackle this problem is through an improved classification procedure. The results from the analyses reported earlier showed that a more accurate prediction is possible. This improved accuracy can be of help in increasing public security, especially for the crimes committed against people. For example, as Bhati (2007, p.25) suggested, an increase of a prison term by one percent can result in a roughly proportional increase in the number of crimes against persons averted. If that increase can be applied to the right individuals, the practical gain would be maximized.

Since this study did not start off with a specific theoretical framework, focusing too much on the effect of individual variables may not be an ideal way to proceed. It is practical to bear in mind that since tree-based models are not additive models, the impact of a single variable may not be correctly understood without thinking the model through in its entirety. Unlike least squares regression, the coefficient of a particular variable is not the main concern, but that variable's capacity to effectively separate competing classes is more of a concern in tree-based models.

Which Models to Use?

If the relationship between the predictors and the outcome can be described linearly, a traditionally used linear model (such as linear regression) can be used without resorting to other sophisticated models (James et al., 2013). According to the findings of this study, logistic regression could compete with other models, depending on the problem at hand.

Support vector machines can provide better solutions if the data points in the multidimensional surface can be separated by a hyperplane. If that hyperplane does not exist, it

severely affects the performance of support vector machines. In the case of recidivism, support vector machines provided promising results in terms of sensitivity.

The problem of prediction is heavily dependent on two things: the prediction goal (i.e., what is the goal of prediction?) and the available data at hand. The model selection takes a back seat in the hierarchy of importance because one can obtain fairly-usable solutions even when one does not use the ideal model for the problem at hand. A least squares regression can be perfectly fine for a linearly-solvable problem. The issue is that real-world data are rarely ideally linear, and most of the time models that can reasonably fit non-linear data are needed. With the absence of relevant and useful data, no model can provide a solution as models cannot create wonders by themselves alone. This study relied extensively on criminal history variables, leaving the potential open for more factors including but not limited to psychosocial, emotional, or community-level variables.

It is essentially important to bear in mind that no classification method is universally better than any other. At the end of the day, the improvement will be based on available data and target functions of each model (Bradley, 1997; Hastie et al., 2009). The performance of any tool will depend on the complexity of the classification boundary. Most forecasting tools will have similar accuracy in the case of a simple classification boundary, while with a complex boundary, machine learning algorithms can provide considerable improvement in the accuracy (Berk & Bleich, 2013). There are various factors that can play a role in one's model choice. Yang et al. (2010) reported that logistic regression can be preferred for its consistency and convenience, while neural networks can be of use in substantial sample sizes with a large number of predictors with small effect sizes.

Accuracy, transparency, and interpretability are the main pillars of prediction, but there are tradeoffs among them as a highly transparent model may not be superior in its predictive power (Zeng et al., 2015). How many false positives would one be willing to accept in exchange for obtaining one fewer false negative? Who will decide that threshold? Following Bushway and Smith's logic (2007), the interpretation of a risk prediction exercise in the sentencing world depends on what one wants to accomplish, and this may become complicated as there may be multiple goals, including deterrence (general and specific), retribution, or incapacitation, all of which are dependent on organizational resources (Steffensmeier, Kramer, & Streifel, 1993). For these reasons, the practitioner can identify a range of plausible thresholds (say, a parole board may want to be more aggressive on sex offender recidivism, thus setting the threshold so the cost would be 1 to 10).

There are two rightful concerns in algorithmic decision making. One is hiddenness of algorithmic decision-making (black-box mode of data collection and storing), and the other is opacity, which means potential incomprehensibility to human reasoning (Danaher, 2016). Practitioners may prefer a model that is more intuitive and less opaque, or black box. Black box models are the ones which are not implicit about selecting appropriate variables, weighting their influences and iteration process. Neural networks are definitely powerful, but at the same time, judges, parole boards or prosecutors want to understand what is going on 'under the hood.' Sometimes, this may lead a preference over "interpretable" models compared to "highest-achieving" models. For example, random forests may provide more intuition, or its relatively easy-to-understand tree structure may be a factor in the model selection process.

In short, as Morgan (2005, p.126) puts it, the world is full of "interaction effects." In the case of large data, the number of potential interaction terms to be considered in model specification increases substantially. Therefore, an analyst may not be able to include (or even realize) these effects into the model. In genetics and biology, most of the time, what matters is not the effect of a single gene, rather the combinations of genes as well as interactions with the environment (Morgan, 2005). For this reason, models that can detect non-linear relationships and interactions that are not obvious in the initial stages of the data analysis may be of help in classification purposes in the U.S. criminal justice system.

Is Logistic Regression Enough?

The answer to that question is: It depends. If an analyst has a linearly-solvable problem and relevant features at hand, a simple OLS or logistic regression can do the work. Specifically, if the model can fit appropriately using several additive main effects, traditional tools should be sufficient. In those cases, traditional tools can compete, if not outperform machine learning tools. However, the world is full of interactions, and an analyst may not be able to specify all potential interactions. Furthermore, no social science theory is comprehensive enough to cover all these minimal but existing effects.

Machine learning models shine in the case of using large datasets (terabytes of data). In that vast data, computers can detect numerous minimal effects, which can add up to the overall predictive accuracy substantially (for example, voice or image recognition). In the case of small datasets, these models may not reach the optimum efficiency.

Another potential reason why logistic regression yielded comparable results is that the current study only relied on dichotomous variables. This may restrict the models due to lack of

information, which other machine learning models could extract. For example, random forests can partition a binary variable into yes or no (as in Search algorithm), but in the case of a continuous variable, there could be many split points. Also, models with binary variables are more shielded against outliers as all values are either zero or one, which is definitely an advantageous situation for logistic regression.

Should We Include Race, Ethnicity or Sex in Risk Assessment?

Sex is a conceptually and technically different factor than race. It would be wrong to argue that psychological sensitivities exist between different races in a prison environment, but the same nuance can be true for sex. For example, in a study examining the risks and needs of status offenders regarding gender and ethnicity, Gavazzi, Yarcheck, and Lim, (2005) found that females displayed different risks and needs than males across various domains. Ignoring female needs in prison settings may diminish rehabilitation efforts and consequentially increase recidivism. For these reasons, excluding sex from risk assessments for the sake of ethical arguments does not serve any good in terms of priorities of policy-making.

In the current study, race seems to be an important predictor in predicting recidivism. Aside from ethical contra-arguments, the inclusion of race is recommended by some due to its influence on performance (Berk, 2012). Methodologically, race is important in increasing accuracy, but that does not by itself justify its inclusion. For example, as social scientists, we can design an unethical experiment that can provide exclusive illumination on a long-lasting research inquiry, but this does not justify its unethical side.

That said, can one control everything even slightly associated with race? The inclusion of neighborhoods as a predictor would be indirectly adding a partial race component. For example,

Berk (2012) convincingly argued that neighborhoods as predictors may be highly correlated with the race due to segregated living in some cities. Lastly, unfortunately, in the case of a limited number of variables, the practitioner may have to include basic demographics because they may not want to lose their contributing effect in the performance.

Instead, race and all other demographics can (and should) be included in risk assessment tools on detecting the needs of the inmates or parolees. Similarly, Andrews and Bonta (2010) classify demographics under the responsivity component in their RNR system or as Hannah-Moffat (2009) argued, demographics can have bearing on responsivity. The treatment and rehabilitation needs of females needs in prison can be categorically different than males, or certain racial groups' vulnerabilities in probation or parole can be unique. In that sense, demographics can help identification of those needs. Incarceration systems can reduce recidivism by focusing on other dimensions such as responsivity and need.

Implementation

One of the most challenging aspects of creating data-informed decision-making tools in the criminal justice system is the implementation. Berk (2012) argued that this procedure will require the talents and cooperation of many parties. To implement principled and systematic application within the criminal justice setting will also require training certain personnel to use and produce visual outputs for the ease of interpretation, especially for non-technical decision-makers.

Considering the fact that many agencies do not have staff trained for this kind of analyses nor financial resources to outsource, these are certain challenges but these are not insurmountable. Consumable data products—obtained through automation—will solve the

problem of trained personnel, but at the end of the day, a data analyst will be needed to interpret and present results in a routine manner.

Limitations

In the current study, offender criminal histories were extensively used in addition to basic demographics. Needless to say, there were other factors that matter and were not included. For example, the communities where offenders return is an important factor affecting the likelihood of recidivism (Taxman et al., 2013). Similarly, none of the substance abuse-related variables or psychological factors (intelligence and impulsivity) were included. If variables like these were available, it is possible that an even more higher accuracy could be obtained.

Official data use can sometimes be inaccurate, and this is certainly true for official records which do not detect all instances of law-breaking. The whole data generating process is a result of a concerted effort between federal- and state-level institutions. It is likely that errors could leak in the data. Lastly, there were many missing cases that were deleted. For example, disciplinary records of the inmates or substance use therapy information were not included in the data. A more comprehensive dataset could result in better predictions.

Future Directions

Specific recidivism (a larcenist who steals again) and general recidivism should be better understood (Liem, 2013). In line with this, an offense-type recidivism classification can be useful. Technically, this would be a multiclass classification, e.g., predicting violent recidivism versus property crime recidivism versus sexual offense recidivism and so on. Neural networks are quite capable of handling these types of classification tasks.

Regardless of domain, risk should be evaluated based on (1) the probability of occurrence and (2) the expected costs of a given outcome (Berk, 2012). Failing to make the distinction of crime type may hide the true cost of false prediction as the consequences of property crime and violent crime is different in nature. Therefore, in addition to recognizing the differential costs existing between false negatives and false positives, and identifying differential consequences of making a false prediction of the crime type should be examined (Berk, 2012).

Some scholars have argued that an encompassing, generalizable recidivism simulation model reflecting national offender population that experienced the criminal justice pipeline through pretrial, probation, jail, prison, and parole would be useful (Ainsworth & Taxman, 2013). It is also important to note here that one may need a simulation tool for each of these specific phases of the pipeline. This empirical problem is yet to be studied.

Since any new offense or technical violation can be considered as a continuation of offending, not all intervention types are expected to affect all types of offending for all types of offenders (Taxman et al., 2013). The RNR framework was designed to remedy this problem and it is generally well-received by the criminological community. Data-driven solutions may increase all three dimensions of the RNR framework by identifying who needs what in a better way.

There are four essential tasks in community supervision (Rhodes, 2011; Taxman, 2008). The first task is to identify risk levels among offenders. The second task is to optimize the available resources to use for supervision. This means that high-risk offenders can be given higher amounts of resources. These resources include anything that is used in supervising offenders (financial, temporal, or manpower costs). The third task is using risk and protective

factors to correctly classify offenders based on their risk levels, while the last task is to implement this overall plan (Rhodes, 2011; Taxman, 2008). After probation or parole, individuals who are released into the communities can be classified depending on their risk and needs. Considering the community supervision system chronic problems, such as insufficient supplies both in financial and human resources, the increasing size of the supervised community is likely to increase the caseloads and decrease the quality of services, resulting in a malfunction in the supervision system (Austin, 2010). To avoid this, probation supervision can be adjusted in content and intensity (Berk, 2012). Accurate prediction tools can allow probation officers to adjust their time and energy on the high-risk offender group suggested by actuarial analysis. Therefore, a better prediction is not only needed for prison release but also in various domains of the criminal justice system, especially considering the reentry problem in the U.S.

According to Kuhn (1962), when a particular school of thought clashes with a newer school of thought and lose the competition, a paradigm shift naturally occurs. Similar to this, we have begun experiencing a slow but a growing clash between traditional statistical analyses and novel estimation techniques. It is important to bear in mind that the data will take many different forms that will be available for the use of social scientists. As science evolves, as the data, methods, and tools get better, social science, in particular, should benefit from these advancements. Forecasting and data-driven solutions are increasingly gaining more popularity in social sciences. As Berk (2008) puts it: “But forecasting also has an applied side because, for a host of economic and policy reasons, being able to anticipate the future with a level of skill that is even a little better than chance can be enormously beneficial” (p.220). Bushway's (2013)

warned that criminology should invest more in the state-of-art tools to keep pace with the science and inform policy better in general. Hopefully, the current analysis is a step in that direction.

APPENDIX A

PAST OFFENSE TYPES

The BJS documented first most serious charge (26 levels) at first arrest (see the BJS publication Recidivism of Prisoners Released in 1994 for the definition of most serious). For a given cycle, prior offenses are regrouped (using the variable A001OFF1 in the Recidivism of Prisoners Released in 1994) into other offense, larceny, burglary, aggravated assault, other sexual, drug possession, robbery, drug trafficking, fraud, motor vehicle theft, weapon-related, other drug-related, rape, other property, other violence, stolen prop, DUI, homicide, arson, manslaughter, and kidnap.

APPENDIX B

CURRENT OFFENSE TYPES (BJS DETAILED OFFENSE CODE)

Murder: Murder of a police officer, murder, attempted murder, conspiracy murder, and unspecified homicide.

Manslaughter: Voluntary manslaughter, vehicular manslaughter, attempted vehicular manslaughter, conspiracy vehicular manslaughter, negligent manslaughter, attempted manslaughter, conspiracy manslaughter and unspecified manslaughter.

Kidnapping: Kidnaping, attempted kidnaping, and conspiracy kidnaping.

Rape: Rape, assault with intent to rape, conspiracy to rape, statutory rape, attempted statutory rape, and conspiracy statutory rape.

Other Sexual Offenses: Incest, sexual abuse-assault, attempted sexual abuse, conspiracy sexual abuse, child molest, attempted child molest, and conspiracy child molest.

Robbery: Armed robbery, attempted armed robbery, conspiracy armed robbery, robbery unspecified, unarmed robbery, attempted unarmed robbery, and conspiracy unarmed robbery.

Other Violence: Forcible sodomy, attempted forcible sodomy, conspiracy forcible sodomy, simple assault, attempted simple assault, conspiracy simple assault, assault public officer, attempted assault public officer, and conspiracy assault public officer, child abuse, attempted child abuse, conspiracy child abuse, violent offense other, arson, and attempted/conspiracy arson.

Other Crimes: Blackmail, attempted blackmail, conspiracy blackmail, hit and run, attempted hit and run, and conspiracy hit and run, embezzlement, attempted embezzlement, conspiracy embezzlement, stolen property receiving, attempted stolen property receiving, conspiracy

stolen property receiving, stolen property traffic, attempted stolen property traffic, conspiracy stolen property traffic, property destruction, attempted property destruction, conspiracy property destruction, hit & run property damage, trespass, attempted trespass, conspiracy trespass, other property, attempted other property, conspiracy other property, escape, attempted escape, conspiracy escape, flight to avoid, attempted flight to avoid, conspiracy flight to avoid, weapon offense, attempted weapon offense, conspiracy weapon offense, parole violation, probation violation, riot, attempted riot, conspiracy riot, habitual offender, contempt of court, court offense, attempted court offense, conspiracy court offense, family related offense, drunk-vagrant-disorderly, morals offense, attempted morals offense, conspiracy morals offense, immigration offense, obstruction justice, attempted obstruction justice, conspiracy obstruction justice, invasion of privacy, commercialized vice, contributing to delinquency, liquor law violation, other public order, attempted other public order, conspiracy other public order, bribery, attempted bribery, conspiracy bribery, juvenile offense, felony unspecified, attempted felony unspecified, conspiracy felony unspecified, misdemeanor, other-unknown, federal embezzlement, federal fraud, federal forgery, federal counterfeit, federal regulatory, federal tax viol., and federal racketeering.

Aggravated Assault: Aggravated assault, attempted aggravated assault, and conspiracy-aggravated assault.

Burglary: Burglary, attempted burglary, conspiracy burglary, burglary implements, attempted to possess burglary tools, and conspiracy to possess burglary tools.

Motor Vehicle Theft or Larceny: Motor vehicle theft, attempted motor vehicle theft, conspiracy motor vehicle theft, attempted grand larceny, conspiracy grand larceny, attempted

petty larceny, conspiracy petty larceny, attempted larceny unknown, conspiracy larceny unknown, unauthorized use vehicle, attempted unauthorized use vehicle, and conspiracy unauthorized use vehicle.

Fraud: Fraud, attempted fraud, and conspiracy fraud.

Drug Trafficking: Traffic heroin, attempted traffic heroin, conspiracy traffic heroin, traffic controlled substance, attempted traffic controlled substance, conspiracy traffic controlled substance, traffic marijuana-hashish, attempted traffic marijuana-hashish, conspiracy traffic marijuana-hashish, traffic drug unspecified, attempted traffic drug unspecified, conspiracy traffic drug unspecified, and traffic cocaine**.**

Drug Possession: Possession heroin, attempted possession heroin, conspiracy possession heroin, possession controlled substance, attempted possession controlled substance, conspiracy possession controlled substance, possession marijuana-hashish, attempted possession marijuana-hashish, conspiracy possession marijuana-hashish, possession drug unspecified, possession cocaine, unspecified heroin, controlled substance unspecified, marijuana-hashish offense unspecified, cocaine offense unspecified, and drug offense unspecified

Traffic-Related: Minor traffic offense, Dwi, DUI, DUI-drugs.

APPENDIX C

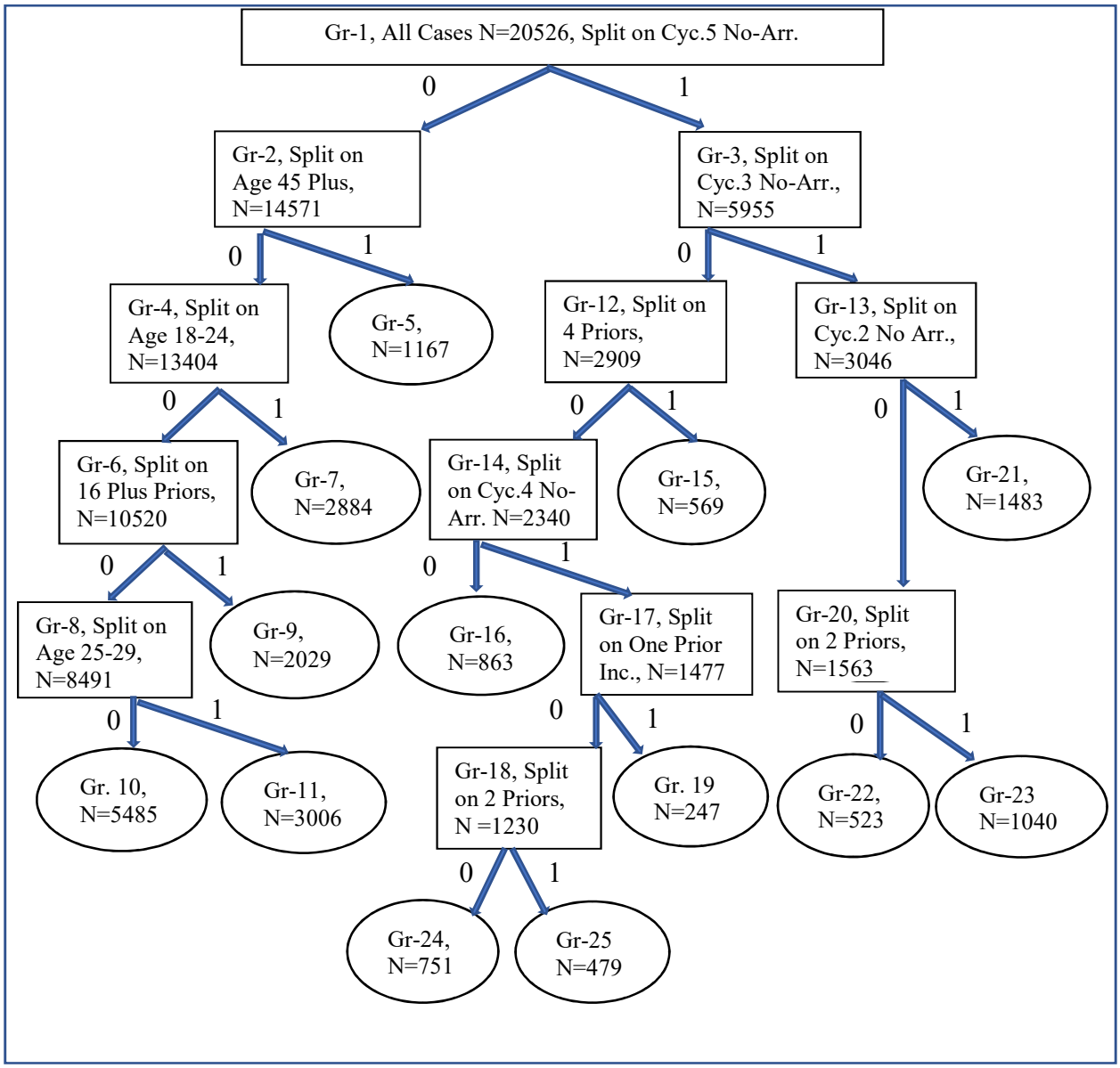
SEARCH SPLIT SUMMARY TABLE

Group No	N	Mean	Var.	Split on	Split into	Outcome
Group 1	20526	0.583	0.243	Cyc.5 No-Arr.	Gr. 2 & 3	-
Group 2	14571	0.719	0.201	Age 45 Plus	Gr. 4 & 5	-
Group 3	5955	0.251	0.188	Cyc.3 No-Arr.	Gr. 12 & 13	-
Group 4	13404	0.739	0.192	Age 18-24	Gr. 6 & 7	-
Group 5	**1167**	**0.490**	**0.250**	-	-	**Non-Recid**.
Group 6	10520	0.711	0.205	16 Plus Priors	Gr. 8 & 9	-
Group 7	**2884**	**0.840**	**0.134**	**3rd Prior Incarcer.**	-	**Recidivist**
Group 8	8491	0. 685	0.215	Age 25-29	Gr. 10 & 11	-
Group 9	**2029**	**0.817**	**0.149**	-	-	**Recidivist**
Group 10	**5485**	**0.644**	**0.229**	-	**Gr. 12 & 13**	**Recidivist**
Group 11	**3006**	**0.762**	**0.181**	-	**Gr. 18 & 19**	**Recidivist**
Group 12	2909	0.390	0.238	Four Priors	Gr. 14 & 15	-
Group 13	3046	0.119	0.105	Cyc.2 No-Arr.	Gr. 20 & 21	-
Group 14	2340	0.475	0.249	Cyc.4 No-Arr.	Gr. 16 & 17	-
Group 15	**569**	**0.040**	**0.038**	-	-	**Non-Recid.**
Group 16	**863**	**0.735**	**0.194**	-	-	**Recidivist**
Group 17	1477	0.323	0.219	One Prior	Gr. 18 & 19	-
Group 18	1230	0.236	0.180	Two Priors	Gr. 24 & 25	-
Group 19	**247**	**0.757**	**0.184**	-	-	**Recidivist**
Group 20	1563	0.217	0.170	Two Priors	Gr. 22 & 23	-
Group 21	**1483**	**0.017**	**0.017**	-	-	**Non-Recid.**
Group 22	**523**	**0.615**	**0.237**	-	-	**Recidivist**
Group 23	**1040**	**0.017**	**0.017**	-	-	**Non-Recid.**
Group 24	**751**	**0.002**	**0.002**	-	-	**Non-Recid.**
Group 25	**479**	**0.603**	**0.239**	-	-	**Recidivist**

Note: These results are based on the mean analysis (Variance Explained by 33.9%).

APPENDIX D

MOST USEFUL SPLITS BY THE SEARCH ALGORITHM



Note: Circles are the final nodes which include a decision (recidivist versus non-recidivist).

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